Mixed Effects Models: Co-creation and Learning Outcomes

BuildBots Frame Paper Analysis

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Overview

This analysis examines the relationship between co-creation level (during LLM interaction) and learning outcomes using mixed effects models. We account for the nested structure of students within groups.

Research Questions:

1. Does co-creation level predict post-test knowledge (T7)?
2. Does baseline knowledge (T1) predict co-creation level (selection effect)?
3. How does knowledge evolve over time (T1 → T4 → T7)?

4. Does the co-creation effect vary by timepoint?

Load Libraries

```
library(tidyverse)      # Data manipulation and visualization
library(lme4)           # Mixed effects models
library(lmerTest)       # p-values for mixed models
library(sjPlot)         # Model visualization
library(performance)    # Model diagnostics
library(ggplot2)        # Advanced plotting
library(gridExtra)      # Multiple plots
library(knitr)          # Tables
library(broom.mixed)    # Tidy model outputs
```

Load and Prepare Data

Load Raw Data

```
# Load children knowledge data (T1, T4, T7)
df_children <- read_csv(
  '../Data/Qualtrics/BuildBots_children_T1T4T7.csv',
  show_col_types = FALSE
)

# Load engagement data to get Group assignments
df_engagement <- read_csv(
  '../Data/Qualtrics/BuildBots_engagement_children_T1-T7.csv',
  show_col_types = FALSE
)

# Load group-level co-creation data
df_cocreation <- read_csv(
  '../Data/BuildbotAnalysis - LLM analysis.csv',
  skip = 1,
  show_col_types = FALSE
)

# Display structure
cat("Children data dimensions:", dim(df_children), "\n")
```

```
## Children data dimensions: 85 95
```

```
cat("Engagement data dimensions:", dim(df_engagement), "\n")
```

```
## Engagement data dimensions: 195 34
```

```
cat("Co-creation data dimensions:", dim(df_cocreation), "\n")
```

```
## Co-creation data dimensions: 11 43
```

Extract Group Mapping and Merge Co-creation Data

IMPORTANT: This must be done BEFORE scoring to ensure Group and Co-creation columns are available.

```r
# Get participant to group mapping from engagement data
mapping <- df_engagement %>%
  select(`Participant ID`, Group) %>%
  distinct() %>%
  mutate(
    `Participant ID` = as.numeric(`Participant ID`),
    Group = as.numeric(Group)
  ) %>%
  filter(!is.na(`Participant ID`), !is.na(Group))

cat("Group mapping extracted:\n")
```

```
## Group mapping extracted:
```

```r
cat("  Unique students:", n_distinct(mapping$`Participant ID`), "\n")
```

```
##   Unique students: 31
```

```r
cat("  Unique groups:", n_distinct(mapping$Group), "\n\n")
```

```
##   Unique groups: 10
```

```r
# Clean co-creation column name
cocreation_col <- "Co-creation Level (0:LLM did it all, 1: concepts by students/rest llm, 2: concepts b

# Prepare co-creation data
df_cocreation_clean <- df_cocreation %>%
  select(Group, Cocreation_Level = all_of(cocreation_col)) %>%
  filter(!is.na(Group), !is.na(Cocreation_Level)) %>%
  mutate(Group = as.numeric(Group))

cat("Co-creation data loaded:\n")
```

```
## Co-creation data loaded:
```

```r
cat("  Groups with co-creation data:", nrow(df_cocreation_clean), "\n\n")
```

```
##   Groups with co-creation data: 9
```

```r
# Merge children data with group mapping, then with co-creation

df_children <- df_children %>%
  mutate(`Participant ID` = as.numeric(`Participant ID`)) %>%
  left_join(mapping, by = "Participant ID") %>%
  filter(!is.na(Group)) %>%
  left_join(df_cocreation_clean, by = "Group") %>%
  filter(!is.na(Cocreation_Level))

cat("Student-level data with co-creation:\n")
```

```
## Student-level data with co-creation:
```

```r
cat("  Total observations:", nrow(df_children), "\n")
```

```
##   Total observations: 74
```

```r
cat("  Unique students:", n_distinct(df_children$`Participant ID`), "\n")
```

```
##   Unique students: 27
```

```r
cat("  Unique groups:", n_distinct(df_children$Group), "\n\n")
```

```
##   Unique groups: 9
```

```r
# Display group sizes
df_children %>%
  group_by(Group, Timepoint) %>%
  summarise(n_students = n(), .groups = 'drop') %>%
  pivot_wider(names_from = Timepoint, values_from = n_students) %>%
  kable(caption = "Number of Students per Group by Timepoint")
```

Table 1: Number of Students per Group by Timepoint

Group	T1	T4	T7
2	3	2	2
3	3	3	3
4	3	3	3
5	3	3	3
6	3	3	3
7	3	3	2
8	2	3	2
9	3	2	2
10	3	3	3

Data Validation and Cleaning

Now check the merged data for any coding issues:

```r
# Check raw values for microcontroller questions
cat("Raw microcontrollers_1 values:\n")
```

```
## Raw microcontrollers_1 values:
```

```r
table(df_children$microcontrollers_1, useNA = "ifany")
```

```
##
##    0    1   99 <NA>
##   25   25   23    1
```

```r
cat("\nRaw microcontrollers_2 values:\n")
```

```
##
## Raw microcontrollers_2 values:
```

```r
table(df_children$microcontrollers_2, useNA = "ifany")
```

```
##
##    0    1   99 <NA>
##   25   21   27    1
```

```r
cat("\nNote: Value 99 = 'I don't know' (no answer attempted)\n")
```

```
##
## Note: Value 99 = 'I don't know' (no answer attempted)
```

```r
cat("Value 1 = correct answer\n")
```

```
## Value 1 = correct answer
```

```r
cat("Value 0 = incorrect answer\n")
```

```
## Value 0 = incorrect answer
```

Create Knowledge Scores

We focus on microcontroller questions as they were the topic of the LLM interaction.

Scoring rules: - Correct answer (value = 1) → 1 point - Incorrect answer (value = 0) → 0 points - "I don't know" (value = 99) → 0 points (treated as incorrect) - Total score range: 0-2 points

Note: We treat "I don't know" (99) as incorrect (0 points) because it indicates lack of knowledge, similar to a wrong answer.

```r
# First, let's check for any unexpected values
cat("Checking for unexpected values:\n")
```

```
## Checking for unexpected values:
```

```r
cat("microcontrollers_1 unique values:",
    unique(as.numeric(df_children$microcontrollers_1)), "\n")
```

```
## microcontrollers_1 unique values: 99 0 1 NA
```

```r
cat("microcontrollers_2 unique values:",
    unique(as.numeric(df_children$microcontrollers_2)), "\n\n")
```

```
## microcontrollers_2 unique values: 0 99 1 NA
```

```r
# Score microcontroller questions
# Expected values: 1 = correct, 0 = incorrect, 99 = "I don't know"
df_children <- df_children %>%
  mutate(
    # Convert to numeric and check for valid values
    mc1_raw = as.numeric(microcontrollers_1),
    mc2_raw = as.numeric(microcontrollers_2),

    # Validate: only 0, 1, or 99 are valid
    mc1_valid = mc1_raw %in% c(0, 1, 99),
    mc2_valid = mc2_raw %in% c(0, 1, 99),

    # Score each question: 1 if correct, 0 if incorrect or "don't know"
    mc1_score = if_else(mc1_valid & mc1_raw == 1, 1, 0),
    mc2_score = if_else(mc2_valid & mc2_raw == 1, 1, 0),

    # Total score (0-2)
    score = mc1_score + mc2_score,

    # Track response types separately for analysis
    n_correct  = if_else(mc1_valid & mc1_raw == 1, 1, 0) +
                 if_else(mc2_valid & mc2_raw == 1, 1, 0),
    n_dont_know = if_else(mc1_valid & mc1_raw == 99, 1, 0) +
                  if_else(mc2_valid & mc2_raw == 99, 1, 0),
    n_incorrect = if_else(mc1_valid & mc1_raw == 0, 1, 0) +
                  if_else(mc2_valid & mc2_raw == 0, 1, 0),

    # Flag invalid responses
    has_invalid = !mc1_valid | !mc2_valid
```

```r
  )

# Report any invalid data
invalid_count <- sum(df_children$has_invalid, na.rm = TRUE)
if (invalid_count > 0) {
  cat("\n[WARNING] Found", invalid_count, "rows with invalid values:\n")
  df_children %>%
    filter(has_invalid) %>%
    select(`Participant ID`, Timepoint, mc1_raw, mc2_raw) %>%
    print()
  cat("\nThese rows will be excluded from analysis.\n\n")
}
```

```
##
## [WARNING] Found 1 rows with invalid values:
## # A tibble: 1 x 4
##   `Participant ID` Timepoint mc1_raw mc2_raw
##              <dbl> <chr>       <dbl>   <dbl>
## 1               25 T4             NA      NA
##
## These rows will be excluded from analysis.
```

```r
# Keep only valid data
df_children_valid <- df_children %>%
  filter(!has_invalid)

# Check what columns we have - DEBUG
cat("\n=== DEBUG: Column check ===\n")
```

```
##
## === DEBUG: Column check ===
```

```r
cat("Total columns:", length(names(df_children_valid)), "\n")
```

```
## Total columns: 108
```

```r
cat("Columns in validated data:\n")
```

```
## Columns in validated data:
```

```r
print(names(df_children_valid))
```

```
##   [1] "Duration (in seconds)"  "Timepoint"              "Participant ID"
##   [4] "age"                    "Gender"                 "NAO_safe"
##   [7] "NAO_competence"         "NAO_reliable"           "NAO_empathetic"
##  [10] "NAO_scary"              "NAO_uncomfortable"      "NAO_social"
##  [13] "Blossom_safe"           "Blossom_competent"      "Blossom_reliable"
##  [16] "Blossom_empathetic"     "Blossom_scary"          "Blossom_uncomfortable"
##  [19] "Blossom_social"         "Marty_safe"             "Marty_competent"
##  [22] "Marty_reliable"         "Marty_empathetic"       "Marty_scary"
##  [25] "Marty_uncomfortable"    "Marty_social"           "SOAS_1"
##  [28] "SOAS_2"                 "SOAS_3"                 "SOAS_4"
##  [31] "SOAS_5"                 "SOAS_6"                 "SOAS_7"
##  [34] "attachment_1"           "attachment_2"           "attachment_3"
##  [37] "attachment_4r"          "attachment_5r"          "intention_1"
##  [40] "intention_2"            "intention_3"            "intention_4"
##  [43] "MAILS_1"                "MAILS_2"                "MAILS_3"
```

```
##  [46] "MAILS_4"                "MAILS_5"                "MAILS_6"
##  [49] "MAILS_7"                "MAILS_8"                "MAILS_9"
##  [52] "MAILS_10"               "robotparts_1"           "robotparts_2"
##  [55] "sensors_1"              "sensors_2"              "microcontrollers_1"
##  [58] "microcontrollers_2"     "AI_DA09"                "CA12"
##  [61] "CI29"                   "CI31"                   "UA19"
##  [64] "GA10"                   "LLMdesign_1"            "GA13"
##  [67] "EA05"                   "GA04"                   "PR1"
##  [70] "PR2"                    "PR3"                    "PR4"
##  [73] "GA17"                   "GA16"                   "EA11"
##  [76] "LLM_Microcontrollers_1" "LLM_Microcontrollers_2" "LLM_Microcontrollers_3"
##  [79] "LLM_Microcontrollers_4" "LLM_Microcontrollers_5" "LLM_Microcontrollers_6"
##  [82] "LLM_Microcontrollers_7" "LLM_Microcontrollers_8" "googled"
##  [85] "googled_time"           "googled_time_3_TEXT"    "googled_topics"
##  [88] "googled_topics_6_TEXT"  "friends"                "friends_time"
##  [91] "friends_time_3_TEXT"    "friends_who"            "friends_who_4_TEXT"
##  [94] "attentioncheck"         "attentioncheck2"        "Group"
##  [97] "Cocreation_Level"       "mc1_raw"                "mc2_raw"
## [100] "mc1_valid"              "mc2_valid"              "mc1_score"
## [103] "mc2_score"              "score"                  "n_correct"
## [106] "n_dont_know"            "n_incorrect"            "has_invalid"
```

```r
cat("\nHas Group column:", "Group" %in% names(df_children_valid), "\n")
```

```
##
## Has Group column: TRUE
```

```r
cat("Has Participant ID:", "Participant ID" %in% names(df_children_valid), "\n")
```

```
## Has Participant ID: TRUE
```

```r
cat("Has Timepoint:", "Timepoint" %in% names(df_children_valid), "\n")
```

```
## Has Timepoint: TRUE
```

```r
cat("Has Cocreation_Level:", "Cocreation_Level" %in% names(df_children_valid), "\n")
```

```
## Has Cocreation_Level: TRUE
```

```r
cat("========================\n\n")
```

```
## ========================
```

```r
# Use valid data for all subsequent analyses
df_children <- df_children_valid

# Display response patterns
cat("Response Pattern Summary (Valid Data Only):\n")
```

```
## Response Pattern Summary (Valid Data Only):
```

```r
cat("==========================================\n\n")
```

```
## ==========================================
```

```r
response_summary <- df_children %>%
  group_by(Timepoint) %>%
  summarise(
    n_students = n(),
    mean_score = mean(score, na.rm = TRUE),
```

```r
    mean_dont_know = mean(n_dont_know, na.rm = TRUE),
    mean_incorrect = mean(n_incorrect, na.rm = TRUE),
    pct_with_dont_know = mean(n_dont_know > 0) * 100, # What % of students had at least one don't know?
    pct_with_incorrect = mean(n_incorrect > 0) * 100, # What % of students had at least one incorrect?
    pct_with_correct = mean(n_correct > 0) * 100,
    pct_dont_know = sum(n_dont_know, na.rm = TRUE) / sum(n_dont_know + n_incorrect + n_correct, na.rm =
    pct_incorrect = sum(n_incorrect, na.rm = TRUE) / sum(n_dont_know + n_incorrect + n_correct, na.rm =
    pct_correct = sum(n_correct, na.rm = TRUE) / sum(n_dont_know + n_incorrect + n_correct, na.rm = TRU
   )

response_summary %>%
  kable(digits = 2, caption = "Response Patterns by Timepoint")
```

Table 2: Response Patterns by Timepoint

Timepoint	n_students	mean_score	mean_dont_know	mean_incorrect	pct_with_dont_know	pct_with_incorrect	pct_with_correct	pct_dont_know	pct_incorrect	pct_correct
T1	26	0.46	1.00	0.54	69.23	42.31	34.62	50.00	26.92	23.08
T4	24	0.79	0.58	0.62	41.67	54.17	70.83	29.17	31.25	39.58
T7	23	0.65	0.43	0.91	34.78	73.91	52.17	21.74	45.65	32.61

```r
# Score distribution by timepoint
df_children %>%
  group_by(Timepoint) %>%
  summarise(
    n = n(),
    mean_score = mean(score, na.rm = TRUE),
    sd_score = sd(score, na.rm = TRUE),
    min = min(score, na.rm = TRUE),
    max = max(score, na.rm = TRUE)
  ) %>%
  kable(digits = 2, caption = "Knowledge Scores by Timepoint")
```

Table 3: Knowledge Scores by Timepoint

Timepoint	n	mean_score	sd_score	min	max
T1	26	0.46	0.71	0	2
T4	24	0.79	0.59	0	2
T7	23	0.65	0.71	0	2

```r
# Detailed breakdown: Number of students per score
cat("\nScore Distribution:\n")
```

```
##
## Score Distribution:
```

```r
df_children %>%
  group_by(Timepoint, score) %>%
  summarise(n = n(), .groups = 'drop') %>%
  pivot_wider(names_from = score, values_from = n, values_fill = 0) %>%
  kable(caption = "Number of Students by Score and Timepoint")
```

Table 4: Number of Students by Score and Timepoint

Timepoint	0	1	2
T1	17	6	3
T4	7	15	2
T7	11	9	3

```
# Response type breakdown (correct, incorrect, don't know)
cat("\nResponse Type Breakdown (per question):\n")
```

```
##
## Response Type Breakdown (per question):
```

```
df_children %>%
  group_by(Timepoint) %>%
  summarise(
    total_questions = n() * 2,
    n_correct = sum(mc1_score + mc2_score),
    n_incorrect = sum(n_incorrect),
    n_dont_know = sum(n_dont_know),
    pct_correct = (n_correct / total_questions) * 100,
    pct_incorrect = (n_incorrect / total_questions) * 100,
    pct_dont_know = (n_dont_know / total_questions) * 100
  ) %>%
  kable(digits = 1, caption = "Percentage of Each Response Type")
```

Table 5: Percentage of Each Response Type

Timepoint	total_questions	n_correct	n_incorrect	n_dont_know	pct_correct	pct_incorrect	pct_dont_know
T1	52	12	14	26	23.1	26.9	50.0
T4	48	19	15	14	39.6	31.2	29.2
T7	46	15	21	10	32.6	45.7	21.7

Reshape to Long Format (After Validation)

Now that we have clean, validated data, we can reshape for longitudinal analysis:

```
# Create long format using VALIDATED data only
# First verify we have all required columns
required_cols <- c("Participant ID", "Group", "Timepoint", "Cocreation_Level", "score")
missing_cols <- setdiff(required_cols, names(df_children))

if (length(missing_cols) > 0) {
  stop("ERROR: Missing required columns: ", paste(missing_cols, collapse = ", "))
}

cat("Creating long format data...\n")
```

```
## Creating long format data...
```

```
df_long <- df_children %>%
  select(`Participant ID`, Group, Timepoint, Cocreation_Level, score,
         n_dont_know, n_incorrect, mc1_score, mc2_score) %>%
```

```r
  mutate(
    # Convert timepoint to numeric (1, 4, 7)
    Time_numeric = case_when(
      Timepoint == "T1" ~ 1,
      Timepoint == "T4" ~ 4,
      Timepoint == "T7" ~ 7,
      TRUE ~ NA_real_
    ),
    # Factor versions for plotting
    Timepoint_factor = factor(Timepoint, levels = c("T1", "T4", "T7")),
    Group_factor = factor(Group),
    Student_ID = factor(`Participant ID`)
  ) %>%
  filter(!is.na(score), !is.na(Time_numeric))

cat("Longitudinal data structure (VALIDATED):\n")
```

```
## Longitudinal data structure (VALIDATED):
```

```r
cat("==========================================\n")
```

```
## ==========================================
```

```r
cat("Total observations:", nrow(df_long), "\n")
```

```
## Total observations: 73
```

```r
cat("Unique students:", n_distinct(df_long$Student_ID), "\n")
```

```
## Unique students: 27
```

```r
cat("Unique groups:", n_distinct(df_long$Group), "\n")
```

```
## Unique groups: 9
```

```r
cat("\nScore range:", min(df_long$score), "to", max(df_long$score), "\n")
```

```
##
## Score range: 0 to 2
```

```r
cat("Mean score:", round(mean(df_long$score), 2), "\n\n")
```

```
## Mean score: 0.63
```

```r
summary(df_long %>% select(Time_numeric, Cocreation_Level, score, Group))
```

```
##   Time_numeric   Cocreation_Level     score            Group
##  Min.   :1.000   Min.   :0.000    Min.   :0.0000   Min.   : 2.000
##  1st Qu.:1.000   1st Qu.:0.500    1st Qu.:0.0000   1st Qu.: 4.000
##  Median :4.000   Median :0.500    Median :1.0000   Median : 6.000
##  Mean   :3.877   Mean   :1.116    Mean   :0.6301   Mean   : 5.904
##  3rd Qu.:7.000   3rd Qu.:1.000    3rd Qu.:1.0000   3rd Qu.: 8.000
##  Max.   :7.000   Max.   :3.000    Max.   :2.0000   Max.   :10.000
```

Descriptive Statistics

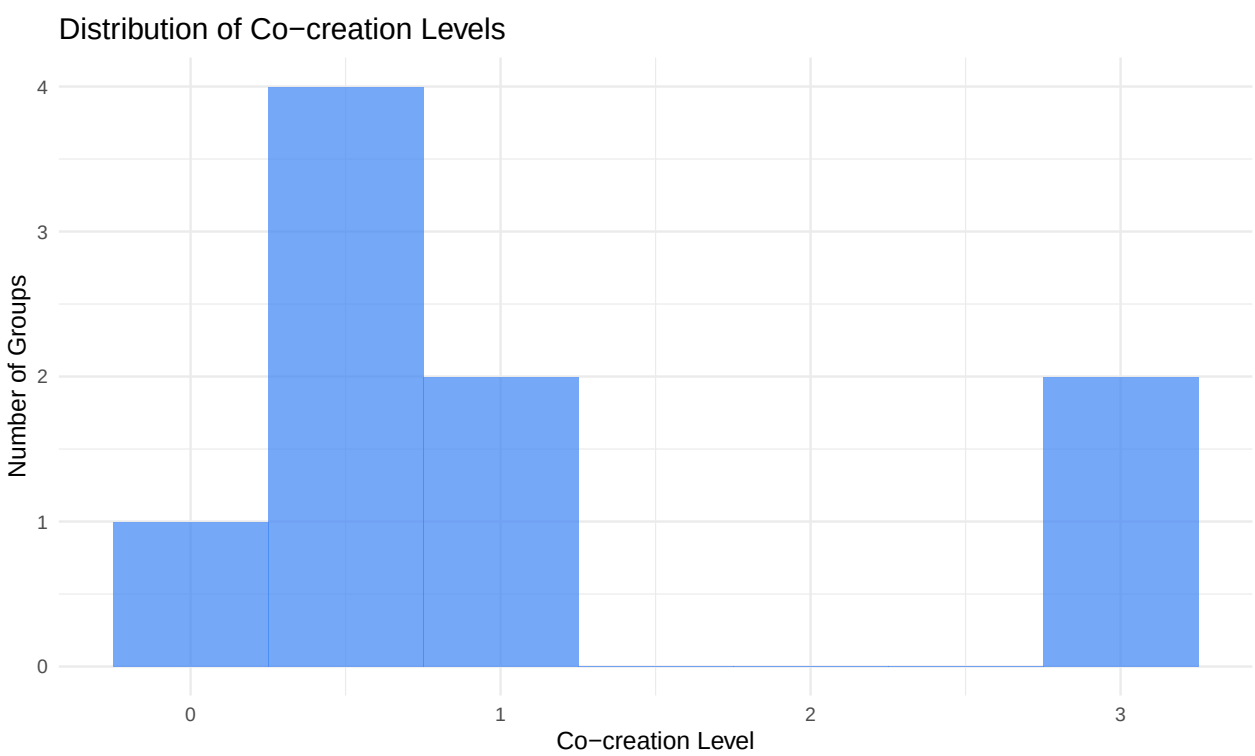
Co-creation Levels

```
df_cocreation_clean %>%
  summarise(
    n_groups = n(),
    mean_level = mean(Cocreation_Level),
    sd_level = sd(Cocreation_Level),
    min = min(Cocreation_Level),
    max = max(Cocreation_Level)
  ) %>%
  kable(digits = 2, caption = "Co-creation Level Distribution")
```

Table 6: Co-creation Level Distribution

n_groups	mean_level	sd_level	min	max
9	1.11	1.11	0	3

```
# Distribution plot
ggplot(df_cocreation_clean, aes(x = Cocreation_Level)) +
  geom_histogram(binwidth = 0.5, fill = "#3b82f6", alpha = 0.7) +
  labs(
    title = "Distribution of Co-creation Levels",
    x = "Co-creation Level",
    y = "Number of Groups"
  ) +
  theme_minimal(base_size = 14)
```



Knowledge Scores Over Time

```r
# Print df_long structure for inspection
cat("\n=== df_long DATA STRUCTURE ===\n")
```

```
##
## === df_long DATA STRUCTURE ===
```

```r
cat("Dimensions:", nrow(df_long), "rows ×", ncol(df_long), "columns\n\n")
```

```
## Dimensions: 73 rows × 13 columns
```

```r
cat("Column names:\n")
```

```
## Column names:
```

```r
print(names(df_long))
```

```
##  [1] "Participant ID"   "Group"            "Timepoint"        "Cocreation_Level"
##  [5] "score"            "n_dont_know"      "n_incorrect"      "mc1_score"
##  [9] "mc2_score"        "Time_numeric"     "Timepoint_factor" "Group_factor"
## [13] "Student_ID"
```

```r
cat("\nFirst 20 rows:\n")
```

```
##
## First 20 rows:
```

```r
print(head(df_long, 20))
```

```
## # A tibble: 20 x 13
##    `Participant ID` Group Timepoint Cocreation_Level score n_dont_know
##               <dbl> <dbl> <chr>                <dbl> <dbl>       <dbl>
##  1                3     2 T1                     3       0           1
##  2                4     3 T1                     1       0           2
##  3                5     4 T1                     1       0           0
##  4                6     3 T1                     1       0           1
##  5                7     4 T1                     1       0           1
##  6                9     5 T1                     0.5     0           2
##  7               10     2 T1                     3       1           0
##  8               11     2 T1                     3       1           1
##  9               12     4 T1                     1       1           1
## 10               13     3 T1                     1       1           1
## 11               14     5 T1                     0.5     0           2
## 12               15     5 T1                     0.5     0           2
## 13               17     7 T1                     0.5     2           0
## 14               18     8 T1                     0.5     0           0
## 15               20     6 T1                     3       1           0
## 16               21     9 T1                     0.5     0           2
## 17               23     7 T1                     0.5     1           1
## 18               24     6 T1                     3       0           1
## 19               25    10 T1                     0       0           1
## 20               26     7 T1                     0.5     2           0
## # i 7 more variables: n_incorrect <dbl>, mc1_score <dbl>, mc2_score <dbl>,
## #   Time_numeric <dbl>, Timepoint_factor <fct>, Group_factor <fct>,
## #   Student_ID <fct>
```

```r
cat("\nSummary statistics:\n")
```

```
##
## Summary statistics:
```

```r
print(summary(df_long))
```

```
##  Participant ID      Group         Timepoint         Cocreation_Level
##  Min.   : 3.00   Min.   : 2.000   Length:73          Min.   :0.000
##  1st Qu.:10.00   1st Qu.: 4.000   Class :character   1st Qu.:0.500
##  Median :18.00   Median : 6.000   Mode  :character   Median :0.500
##  Mean   :17.58   Mean   : 5.904                      Mean   :1.116
##  3rd Qu.:26.00   3rd Qu.: 8.000                      3rd Qu.:1.000
##  Max.   :32.00   Max.   :10.000                      Max.   :3.000
##
##      score         n_dont_know      n_incorrect       mc1_score
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
##  Median :1.0000   Median :0.0000   Median :1.0000   Median :0.0000
##  Mean   :0.6301   Mean   :0.6849   Mean   :0.6849   Mean   :0.3425
##  3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000
##  Max.   :2.0000   Max.   :2.0000   Max.   :2.0000   Max.   :1.0000
##
##    mc2_score       Time_numeric   Timepoint_factor  Group_factor   Student_ID
##  Min.   :0.0000   Min.   :1.000   T1:26            3      : 9    3      : 3
##  1st Qu.:0.0000   1st Qu.:1.000   T4:24            4      : 9    4      : 3
##  Median :0.0000   Median :4.000   T7:23            5      : 9    5      : 3
##  Mean   :0.2877   Mean   :3.877                    6      : 9    6      : 3
##  3rd Qu.:1.0000   3rd Qu.:7.000                    7      : 8    7      : 3
##  Max.   :1.0000   Max.   :7.000                    10     : 8    9      : 3
##                                                    (Other):21    (Other):55
```

```r
cat("\n==============================\n\n")
```

```
##
## ==============================
```

```r
# CHECK FOR OUTLIERS (scores < 0 or > 2)
cat("\n=== CHECKING FOR INVALID SCORES ===\n")
```

```
##
## === CHECKING FOR INVALID SCORES ===
```

```r
outliers_low <- df_long %>% filter(score < 0)
outliers_high <- df_long %>% filter(score > 2)

if (nrow(outliers_low) > 0) {
  cat("\n[WARNING] Found", nrow(outliers_low), "scores BELOW 0:\n")
  print(outliers_low %>% select(`Participant ID`, Group, Timepoint, score, mc1_raw, mc2_raw, mc1_score,
} else {
  cat("\n[OK] No scores below 0\n")
}
```

```
##
## [OK] No scores below 0
```

```r
if (nrow(outliers_high) > 0) {
  cat("\n[WARNING] Found", nrow(outliers_high), "scores ABOVE 2:\n")
  print(outliers_high %>% select(`Participant ID`, Group, Timepoint, score, mc1_raw, mc2_raw, mc1_score
} else {
  cat("\n[OK] No scores above 2\n")
}
```

```
##
## [OK] No scores above 2
```

```r
cat("\nScore range in df_long:", min(df_long$score), "to", max(df_long$score), "\n")
```

```
##
## Score range in df_long: 0 to 2
```

```r
cat("====================================\n\n")
```

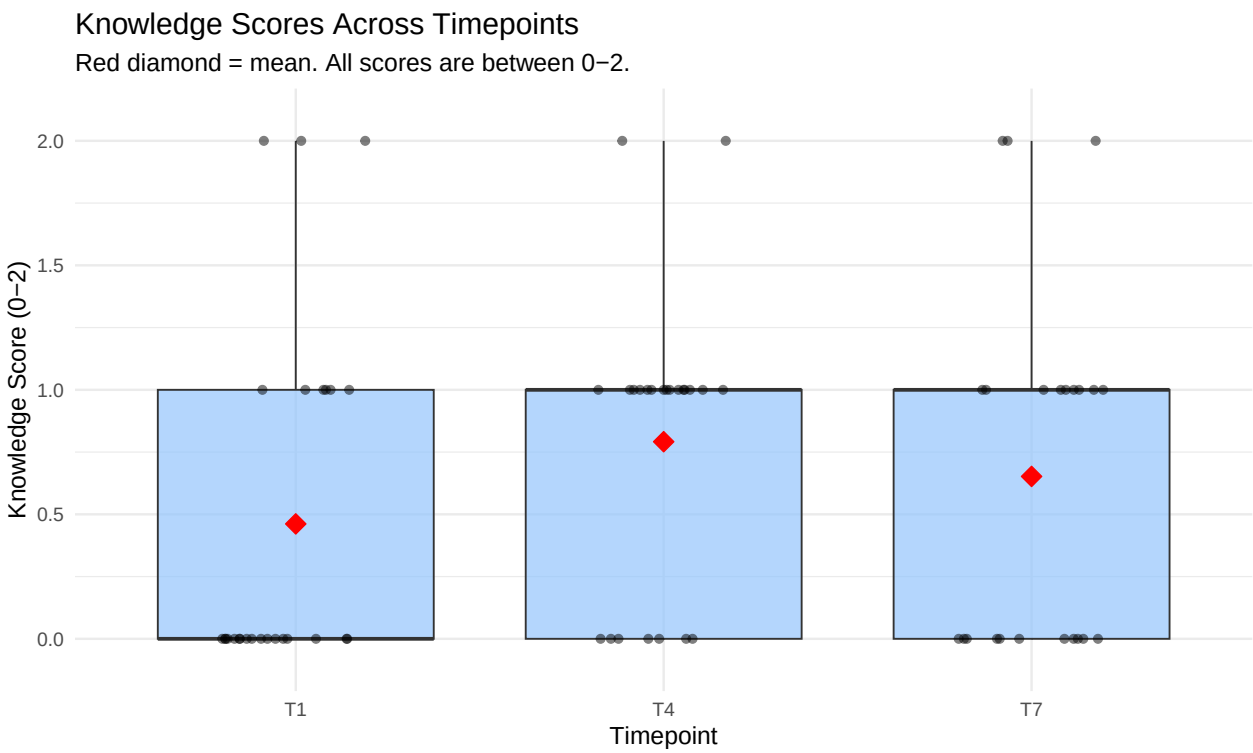
```
## ====================================
```

```r
# Summary by timepoint
df_long %>%
  group_by(Timepoint_factor) %>%
  summarise(
    n_students = n(),
    mean_score = mean(score),
    sd_score = sd(score),
    se_score = sd_score / sqrt(n_students)
  ) %>%
  kable(digits = 2, caption = "Knowledge Scores by Timepoint")
```

Table 7: Knowledge Scores by Timepoint

Timepoint_factor	n_students	mean_score	sd_score	se_score
T1	26	0.46	0.71	0.14
T4	24	0.79	0.59	0.12
T7	23	0.65	0.71	0.15

```r
# Plot knowledge evolution
# NOTE: Using geom_jitter with height=0 to avoid y-axis displacement
ggplot(df_long, aes(x = Timepoint_factor, y = score)) +
  geom_boxplot(fill = "#93C5FD", alpha = 0.7) +
  geom_jitter(width = 0.2, height = 0, alpha = 0.5, size = 2) +  # height=0 prevents y-axis jitter
  stat_summary(fun = mean, geom = "point", shape = 23, size = 4,
               fill = "red", color = "red") +
  scale_y_continuous(limits = c(-0.1, 2.1), breaks = seq(0, 2, 0.5)) +  # Set y-axis limits
  labs(
    title = "Knowledge Scores Across Timepoints",
    subtitle = "Red diamond = mean. All scores are between 0-2.",
    x = "Timepoint",
    y = "Knowledge Score (0-2)"
  ) +
  theme_minimal(base_size = 14)
```



Knowledge by Co-creation Level

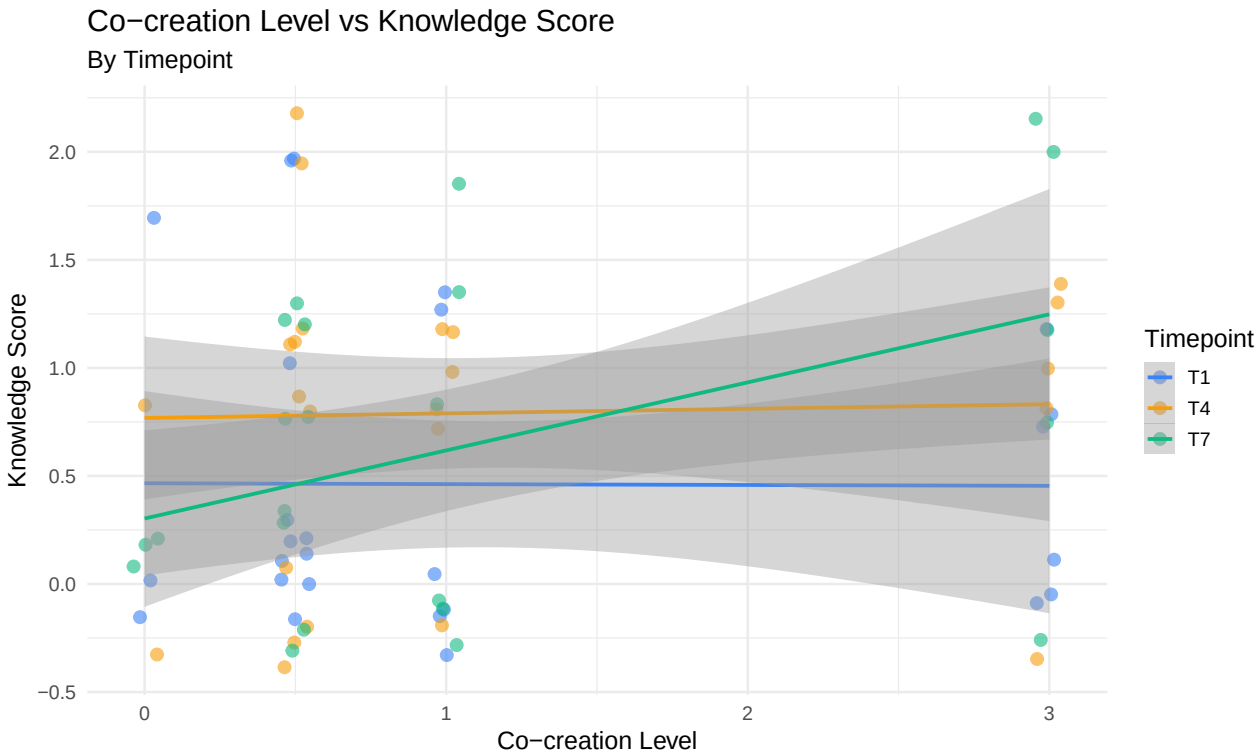
```
df_long %>%
  group_by(Cocreation_Level, Timepoint_factor) %>%
  summarise(
    n = n(),
    mean_score = mean(score),
    sd_score = sd(score),
    .groups = 'drop'
  ) %>%
  pivot_wider(
    names_from = Timepoint_factor,
    values_from = c(n, mean_score, sd_score)
  ) %>%
  kable(digits = 2, caption = "Knowledge Scores by Co-creation Level and Timepoint")
```

Table 8: Knowledge Scores by Co-creation Level and Timepoint

Cocreation_Level	n_T1	n_T4	n_T7	mean_score_T1	mean_score_T4	mean_score_T7	sd_score_T1	sd_score_T4	sd_score_T7
0.0	3	2	3	0.67	0.50	0.00	1.15	0.71	0.00
0.5	11	11	9	0.45	0.82	0.56	0.82	0.75	0.53
1.0	6	6	6	0.33	0.83	0.67	0.52	0.41	0.82
3.0	6	5	5	0.50	0.80	1.20	0.55	0.45	0.84

```
# Plot
ggplot(df_long, aes(x = Cocreation_Level, y = score, color = Timepoint_factor)) +
  geom_point(alpha = 0.6, size = 3, position = position_jitter(width = 0.05)) +
  geom_smooth(method = "lm", se = TRUE) +
```

```r
  scale_color_manual(values = c("T1" = "#3b82f6", "T4" = "#f59e0b", "T7" = "#10b981")) +
  labs(
    title = "Co-creation Level vs Knowledge Score",
    subtitle = "By Timepoint",
    x = "Co-creation Level",
    y = "Knowledge Score",
    color = "Timepoint"
  ) +
  theme_minimal(base_size = 14)
```



Mixed Effects Models

Model 1: T7 Only - Co-creation Predicts Learning

Does co-creation level predict post-test knowledge (T7)?

```r
# This model runs at the student level with group-level random effects.
# Filter T7 data only
df_t7 <- df_long %>% filter(Timepoint == "T7")

# Mixed model with random intercept for groups
model1 <- lmer(
  score ~ Cocreation_Level + (1 | Group_factor),
  data = df_t7,
  REML = TRUE
)

# Summary
summary(model1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: score ~ Cocreation_Level + (1 | Group_factor)
##    Data: df_t7
##
## REML criterion at convergence: 47.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.9348 -0.7137 -0.3845  0.8365  2.1426
##
## Random effects:
##  Groups       Name        Variance Std.Dev.
##  Group_factor (Intercept) 0.0000   0.0000
##  Residual                 0.4161   0.6451
## Number of obs: 23, groups:  Group_factor, 9
##
## Fixed effects:
##                  Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)        0.3029     0.1964 21.0000   1.542   0.1380
## Cocreation_Level   0.3151     0.1291 21.0000   2.441   0.0236 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## Cocretn_Lvl -0.729
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
# Tidy output
tidy(model1, conf.int = TRUE) %>%
  kable(digits = 3, caption = "Model 1: Co-creation → T7 Knowledge")
```

Table 9: Model 1: Co-creation → T7 Knowledge

effect	group	term	estimate	std.error	statistic	df	p.value	conf.low	conf.high
fixed	NA	(Intercept)	0.303	0.196	1.542	21	0.138	-0.106	0.711
fixed	NA	Cocreation_Level	0.315	0.129	2.441	21	0.024	0.047	0.583
ran_pars	Group_factor	sd__(Intercept)	0.000	NA	NA	NA	NA	NA	NA
ran_pars	Residual	sd__Observation	0.645	NA	NA	NA	NA	NA	NA

```r
# Model performance
cat("\nModel Performance:\n")
```

```
##
## Model Performance:
```

```r
performance::icc(model1)
```

```
## [1] NA
```

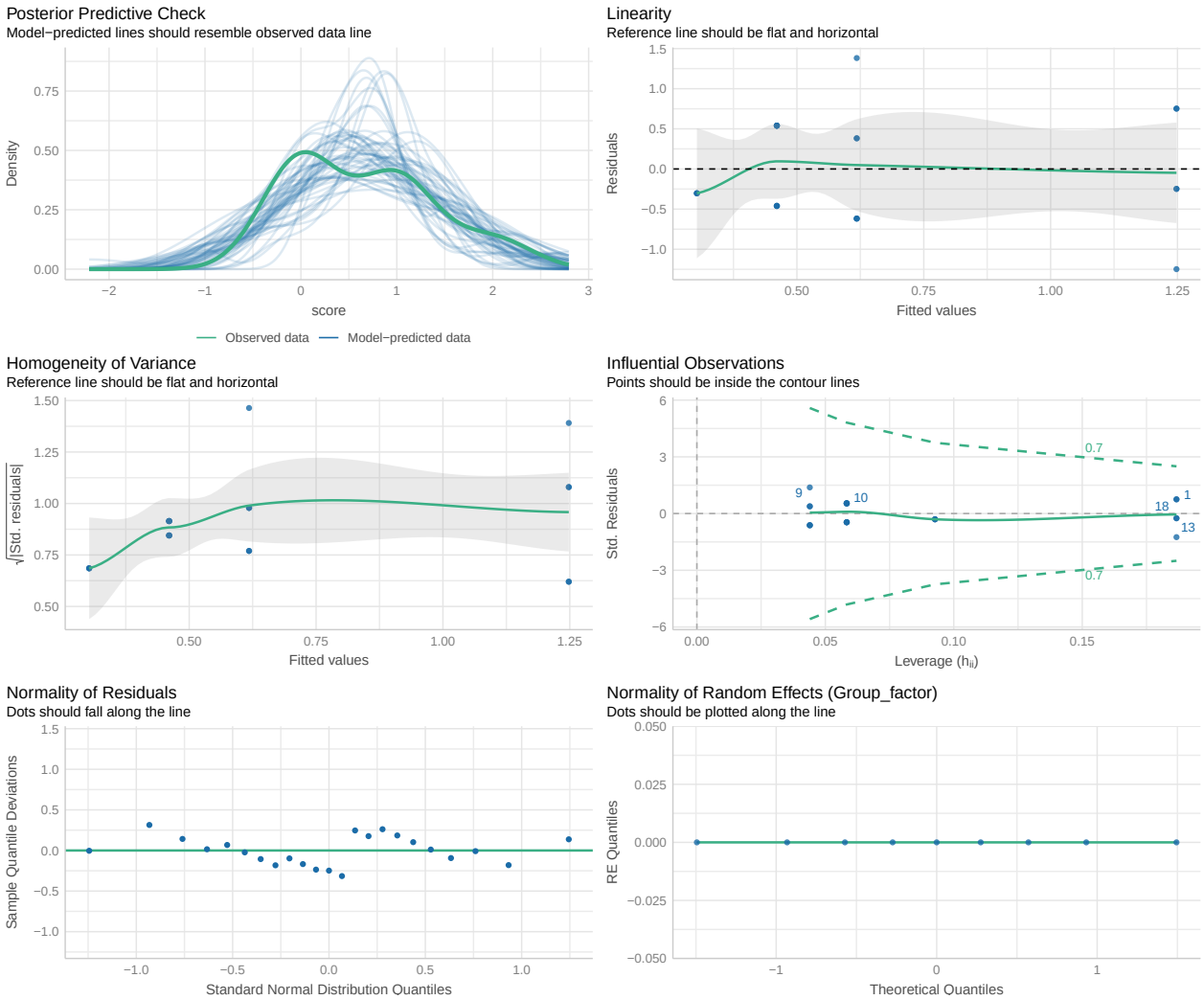
```r
performance::r2(model1)
```

```
## Random effect variances not available. Returned R2 does not account for random effects.
```

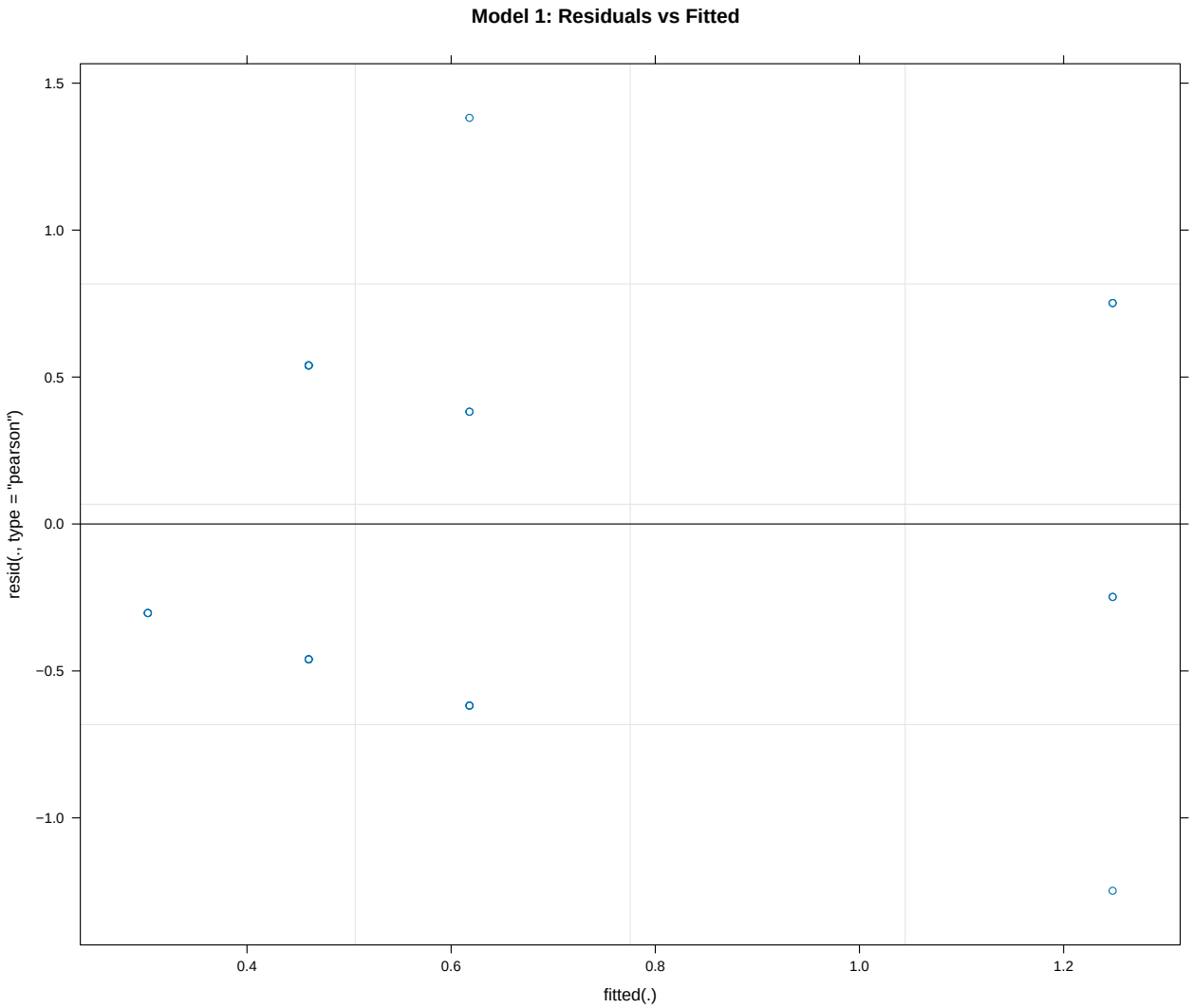
```
## # R2 for Mixed Models
##
##   Conditional R2: NA
##      Marginal R2: 0.213
```

Model 1 Diagnostics

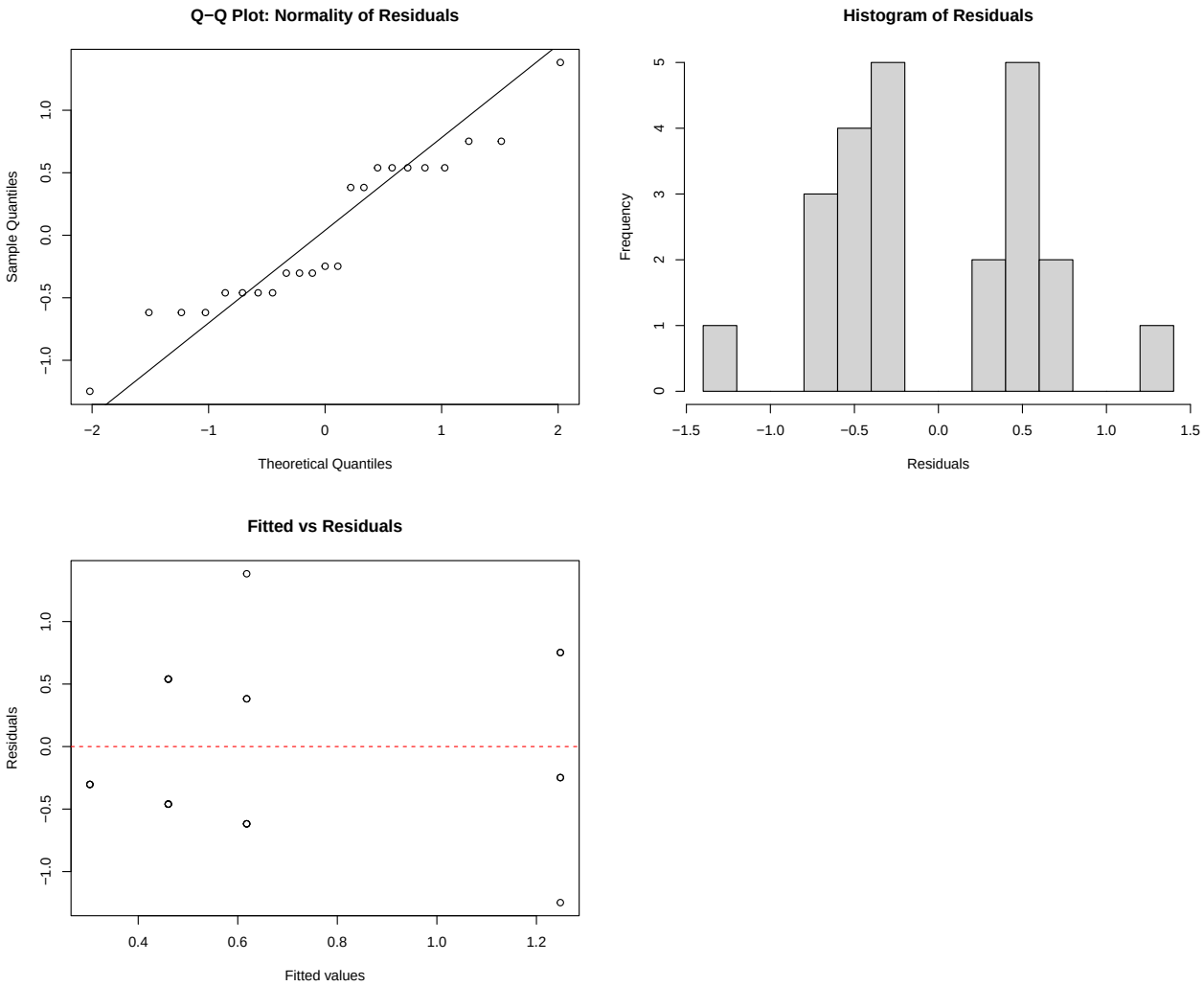
```r
# Check assumptions
check_model(model1, panel = TRUE)
```




```r
# Additional diagnostic plots
par(mfrow = c(2, 2))
plot(model1, main = "Model 1: Residuals vs Fitted")
```




```r
qqnorm(resid(model1), main = "Q-Q Plot: Normality of Residuals")
qqline(resid(model1))
hist(resid(model1), main = "Histogram of Residuals", xlab = "Residuals", breaks = 10)
plot(fitted(model1), resid(model1), main = "Fitted vs Residuals",
     xlab = "Fitted values", ylab = "Residuals")
abline(h = 0, lty = 2, col = "red")
par(mfrow = c(1, 1))
```



Model 1 Interpretation

```
cat("\n=== MODEL 1 INTERPRETATION ===\n")
```

```
##
## === MODEL 1 INTERPRETATION ===
```

```
cat("Research Question: Does co-creation level predict T7 knowledge?\n\n")
```

```
## Research Question: Does co-creation level predict T7 knowledge?
```

```
# Extract key statistics
beta_cocreation <- fixef(model1)["Cocreation_Level"]
se_cocreation <- sqrt(diag(vcov(model1)))["Cocreation_Level"]
t_value <- summary(model1)$coefficients["Cocreation_Level", "t value"]
p_value <- summary(model1)$coefficients["Cocreation_Level", "Pr(>|t|)"]
ci <- confint(model1, parm = "Cocreation_Level", level = 0.95)

cat("Fixed Effects:\n")
```

```
## Fixed Effects:
```

```r
cat(sprintf("  Co-creation coefficient (beta): %.3f\n", beta_cocreation))
```

```
##   Co-creation coefficient (beta): 0.315
```

```r
cat(sprintf("  Standard error: %.3f\n", se_cocreation))
```

```
##   Standard error: 0.129
```

```r
cat(sprintf("  95%% CI: [%.3f, %.3f]\n", ci[1], ci[2]))
```

```
##   95% CI: [0.063, 0.567]
```

```r
cat(sprintf("  t-value: %.3f\n", t_value))
```

```
##   t-value: 2.441
```

```r
cat(sprintf("  p-value: %.4f\n", p_value))
```

```
##   p-value: 0.0236
```

```r
# Effect size interpretation
cat("\nEffect Interpretation:\n")
```

```
##
## Effect Interpretation:
```

```r
if (p_value < 0.05) {
  cat("  [YES] SIGNIFICANT effect of co-creation on T7 knowledge\n")
  cat(sprintf("  - For each 1-unit increase in co-creation level,\n"))
  cat(sprintf("    knowledge score increases by %.2f points (out of 2)\n", beta_cocreation))
  cat(sprintf("  - This represents a %.1f%% increase per co-creation level\n", beta_cocreation/2*100))
} else {
  cat("  [NO] NO significant effect of co-creation on T7 knowledge\n")
}
```

```
##   [YES] SIGNIFICANT effect of co-creation on T7 knowledge
##   - For each 1-unit increase in co-creation level,
##     knowledge score increases by 0.32 points (out of 2)
##   - This represents a 15.8% increase per co-creation level
```

```r
# Random effects
var_group <- as.data.frame(VarCorr(model1))$vcov[1]
var_resid <- attr(VarCorr(model1), "sc")^2
icc_val <- var_group / (var_group + var_resid)

cat("\nRandom Effects:\n")
```

```
##
## Random Effects:
```

```r
cat(sprintf("  Group variance: %.3f\n", var_group))
```

```
##   Group variance: 0.000
```

```r
cat(sprintf("  Residual variance: %.3f\n", var_resid))
```

```
##   Residual variance: 0.416
```

```r
cat(sprintf("  ICC (Intraclass Correlation): %.3f\n", icc_val))
```

```
##   ICC (Intraclass Correlation): 0.000
```

```r
cat(sprintf("  - %.1f%% of variance is between groups\n", icc_val*100))
```

```
##   - 0.0% of variance is between groups
```

```r
cat(sprintf("  - %.1f%% of variance is within groups (between students)\n", (1-icc_val)*100))
```

```
##   - 100.0% of variance is within groups (between students)
```

```r
# Model fit
cat("\nModel Fit:\n")
```

```
##
## Model Fit:
```

```r
cat(sprintf("  AIC: %.1f\n", AIC(model1)))
```

```
##   AIC: 55.5
```

```r
cat(sprintf("  BIC: %.1f\n", BIC(model1)))
```

```
##   BIC: 60.1
```

```r
cat(sprintf("  Log-likelihood: %.2f\n", logLik(model1)))
```

```
##   Log-likelihood: -23.77
```

```r
# R-squared
r2_vals <- r2(model1)
```

```
## Random effect variances not available. Returned R2 does not account for random effects.
```

```r
cat(sprintf("  Marginal R² (fixed effects only): %.3f\n", r2_vals$R2_marginal))
```

```
##   Marginal R² (fixed effects only): 0.213
```

```r
cat(sprintf("  Conditional R² (fixed + random): %.3f\n", r2_vals$R2_conditional))
```

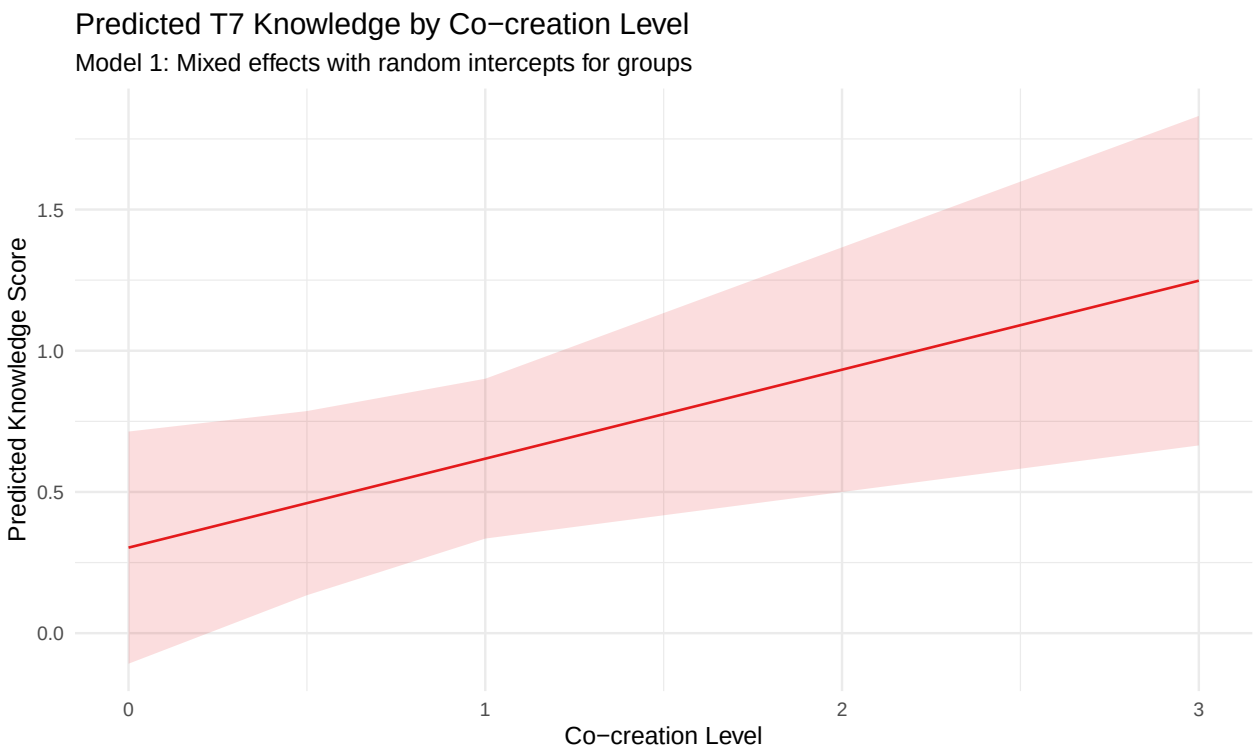
```
##   Conditional R² (fixed + random): NA
```

```r
cat("\n================================\n")
```

```
##
## ================================
```

Model 1 Visualization

```r
# Plot fixed effects
plot_model(model1, type = "pred", terms = "Cocreation_Level") +
  labs(
    title = "Predicted T7 Knowledge by Co-creation Level",
    subtitle = "Model 1: Mixed effects with random intercepts for groups",
    x = "Co-creation Level",
    y = "Predicted Knowledge Score"
  ) +
  theme_minimal(base_size = 14)
```



Model 1 : another approach since group random effects at zero

```
# Load required packages
library(sandwich)
library(lmtest)
library(broom)
library(knitr)

# Filter T7 data only
df_t7 <- df_long %>% filter(Timepoint == "T7")

# Fit simple linear regression
model1_simple <- lm(score ~ Cocreation_Level, data = df_t7)

# Get cluster-robust standard errors
model1_robust <- coeftest(model1_simple, vcov = vcovCL, cluster = ~Group_factor)

# Display results
cat("Linear Regression with Cluster-Robust Standard Errors\n")
```

```
## Linear Regression with Cluster-Robust Standard Errors
```

```
cat("=========================================================\n\n")
```

```
## ========================================================
```

```
print(model1_robust)
```

```
##
## t test of coefficients:
##
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.302872   0.187073  1.6190  0.12037
```

```
## Cocreation_Level 0.315057   0.098718  3.1915  0.00439 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Get confidence intervals with robust SEs
robust_ci <- coefci(model1_simple, vcov = vcovCL, cluster = ~Group_factor)
print(robust_ci)
```

```
##                        2.5 %    97.5 %
## (Intercept)      -0.08616662 0.6919107
## Cocreation_Level  0.10976054 0.5203526
```

```r
# Create a tidy table
robust_results <- tidy(model1_robust, conf.int = TRUE) %>%
  kable(digits = 3, caption = "Model 1: Co-creation → T7 Knowledge (Cluster-Robust SE)")

print(robust_results)
```

```
##
##
## Table: Model 1: Co-creation → T7 Knowledge (Cluster-Robust SE)
##
## |term             | estimate| std.error| statistic| p.value| conf.low| conf.high|
## |:----------------|--------:|---------:|---------:|-------:|--------:|---------:|
## |(Intercept)      |    0.303|     0.187|     1.619|   0.120|   -0.086|     0.692|
## |Cocreation_Level |    0.315|     0.099|     3.191|   0.004|    0.110|     0.520|
```

```r
# Model R-squared (not adjusted for clustering, but informative)
cat("\nModel R-squared:", summary(model1_simple)$r.squared, "\n")
```

```
##
## Model R-squared: 0.2210281
```

```r
cat("Adjusted R-squared:", summary(model1_simple)$adj.r.squared, "\n")
```

```
## Adjusted R-squared: 0.1839342
```

```r
cat("\nNumber of observations:", nobs(model1_simple), "\n")
```

```
##
## Number of observations: 23
```

```r
cat("Number of clusters (groups):", length(unique(df_t7$Group_factor)), "\n")
```

```
## Number of clusters (groups): 9
```

Model 2: T1 Predicts Co-creation (Selection Effect)

2. Does baseline knowledge (T1) predict co-creation level (selection effect)?

```r
# Filter T1 data
df_t1 <- df_long %>% filter(Timepoint == "T1")

# Since co-creation is at group level, we need group-level analysis
df_t1_group <- df_t1 %>%
  group_by(Group_factor) %>%
  summarise(
    mean_T1_score = mean(score, na.rm = TRUE),
    Cocreation_Level = first(Cocreation_Level)
```

```r
  )

# Simple linear model (no random effects needed - one obs per group)
model2 <- lm(Cocreation_Level ~ mean_T1_score, data = df_t1_group)

# Summary
summary(model2)
```

```
##
## Call:
## lm(formula = Cocreation_Level ~ mean_T1_score, data = df_t1_group)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.1167 -0.6000 -0.6000 -0.1083  1.8917
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)     1.1000     0.5315   2.070   0.0773 .
## mean_T1_score   0.0250     0.7973   0.031   0.9759
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.188 on 7 degrees of freedom
## Multiple R-squared:  0.0001404,  Adjusted R-squared:  -0.1427
## F-statistic: 0.0009833 on 1 and 7 DF,  p-value: 0.9759
```

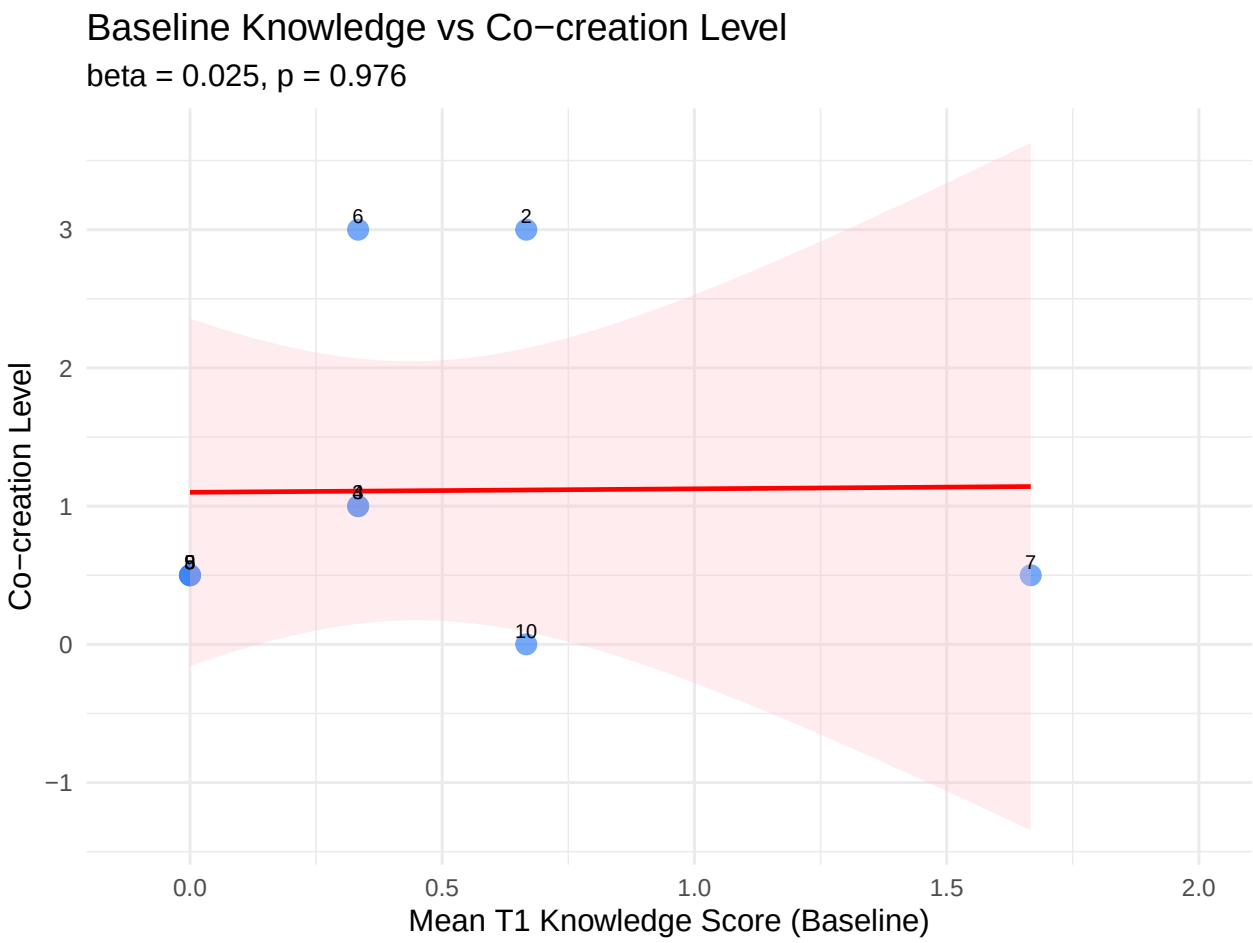
```r
# Tidy output
tidy(model2, conf.int = TRUE) %>%
  kable(digits = 3, caption = "Model 2: T1 Knowledge → Co-creation Level")
```

Table 10: Model 2: T1 Knowledge → Co-creation Level

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1.100	0.532	2.070	0.077	-0.157	2.357
mean_T1_score	0.025	0.797	0.031	0.976	-1.860	1.910

Model 2 Visualization

```r
ggplot(df_t1_group, aes(x = mean_T1_score, y = Cocreation_Level)) +
  geom_point(size = 4, alpha = 0.7, color = "#3b82f6") +
  geom_smooth(method = "lm", se = TRUE, color = "red", fill = "pink", alpha = 0.3) +
  geom_text(aes(label = Group_factor), nudge_y = 0.1, size = 3) +
  labs(
    title = "Baseline Knowledge vs Co-creation Level",
    subtitle = paste0("beta = ", round(coef(model2)[2], 3),
                      ", p = ", round(summary(model2)$coefficients[2, 4], 3)),
    x = "Mean T1 Knowledge Score (Baseline)",
    y = "Co-creation Level"
  ) +
  xlim(-0.1, 2) +
  theme_minimal(base_size = 14)
```



Model 2 Interpretation

```
cat("\n=== MODEL 2 INTERPRETATION ===\n")
```

```
##
## === MODEL 2 INTERPRETATION ===
```

```
cat("Research Question: Does baseline knowledge predict co-creation level?\n")
```

```
## Research Question: Does baseline knowledge predict co-creation level?
```

```
cat("(Testing for selection effects)\n\n")
```

```
## (Testing for selection effects)
```

```
# Extract coefficients
beta_t1 <- coef(model2)["mean_T1_score"]
se_t1 <- summary(model2)$coefficients["mean_T1_score", "Std. Error"]
t_val <- summary(model2)$coefficients["mean_T1_score", "t value"]
p_val <- summary(model2)$coefficients["mean_T1_score", "Pr(>|t|)"]
ci <- confint(model2, "mean_T1_score", level = 0.95)
r_squared <- summary(model2)$r.squared
adj_r_squared <- summary(model2)$adj.r.squared

cat("Regression Results:\n")
```

```
## Regression Results:
```

```r
cat(sprintf("  Coefficient (beta): %.3f\n", beta_t1))
```

```
##   Coefficient (beta): 0.025
```

```r
cat(sprintf("  Standard error: %.3f\n", se_t1))
```

```
##   Standard error: 0.797
```

```r
cat(sprintf("  95%% CI: [%.3f, %.3f]\n", ci[1], ci[2]))
```

```
##   95% CI: [-1.860, 1.910]
```

```r
cat(sprintf("  t(%d) = %.3f\n", model2$df.residual, t_val))
```

```
##   t(7) = 0.031
```

```r
cat(sprintf("  p-value: %.4f\n", p_val))
```

```
##   p-value: 0.9759
```

```r
cat(sprintf("  R-squared = %.3f, Adjusted R-squared = %.3f\n", r_squared, adj_r_squared))
```

```
##   R-squared = 0.000, Adjusted R-squared = -0.143
```

```r
cat("\n=== CRITICAL INTERPRETATION ===\n")
```

```
##
## === CRITICAL INTERPRETATION ===
```

```r
if (p_val < 0.05) {
  cat("  [WARNING] SELECTION EFFECT DETECTED\n\n")
  if (beta_t1 > 0) {
    cat("  -> Groups with HIGHER baseline knowledge had HIGHER co-creation\n")
    cat("  -> This suggests selection bias:\n")
    cat("     - Stronger students may have engaged more actively\n")
    cat("     - Co-creation level may reflect prior ability, not just intervention\n")
  } else {
    cat("  -> Groups with LOWER baseline knowledge had HIGHER co-creation\n")
    cat("  -> This suggests compensatory engagement:\n")
    cat("     - Weaker students may have relied more on LLM\n")
  }
  cat("\n  IMPLICATION FOR CAUSAL INFERENCE:\n")
  cat("  The relationship between co-creation and T7 outcomes may be\n")
  cat("  confounded by baseline differences. Consider:\n")
  cat("  - Adding T1 as a covariate in main models\n")
  cat("  - Reporting selection effect in limitations\n")
  cat("  - Using propensity score matching or ANCOVA\n")
} else {
  cat("  [OK] NO SELECTION EFFECT\n\n")
  cat("  -> Baseline knowledge does NOT predict co-creation level\n")
  cat("  -> Co-creation appears independent of prior knowledge\n")
  cat("  -> This supports causal interpretation:\n")
  cat("     - Co-creation -> Learning relationship is not confounded\n")
  cat("     - by baseline group differences\n")
  cat("\n  IMPLICATION FOR CAUSAL INFERENCE:\n")
  cat("  The co-creation effect on T7 likely represents a genuine\n")
  cat("  learning effect rather than selection artifact.\n")
}
```

```
##   [OK] NO SELECTION EFFECT
##
##   -> Baseline knowledge does NOT predict co-creation level
##   -> Co-creation appears independent of prior knowledge
##   -> This supports causal interpretation:
##      - Co-creation -> Learning relationship is not confounded
##      - by baseline group differences
##
##   IMPLICATION FOR CAUSAL INFERENCE:
##   The co-creation effect on T7 likely represents a genuine
##   learning effect rather than selection artifact.
```

```r
cat("\n================================\n")
```

```
##
## ================================
```

Model 3: Longitudinal - Time

3. How does knowledge evolve over time (T1 → T4 → T7)?

```r
# Mixed model with time
model3 <- lmer(
  score ~ Time_numeric  + (1 | Group_factor) + (1 | Student_ID),
  data = df_long,
  REML = TRUE
)

# Summary
summary(model3)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: score ~ Time_numeric + (1 | Group_factor) + (1 | Student_ID)
##    Data: df_long
##
## REML criterion at convergence: 156.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.0849 -0.7893  0.3926  0.5404  2.1657
##
## Random effects:
##  Groups       Name        Variance Std.Dev.
##  Student_ID   (Intercept) 0.0000   0.0000
##  Group_factor (Intercept) 0.0000   0.0000
##  Residual                 0.4581   0.6768
## Number of obs: 73, groups:  Student_ID, 27; Group_factor, 9
##
## Fixed effects:
##              Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)   0.50084    0.14807 71.00000   3.382  0.00117 **
## Time_numeric  0.03335    0.03227 71.00000   1.034  0.30487
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##             (Intr)
## Time_numerc -0.845
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
# Tidy output
tidy(model3, conf.int = TRUE) %>%
  kable(digits = 3, caption = "Model 3: Knowledge over Time")
```

Table 11: Model 3: Knowledge over Time

effect	group	term	estimate	std.error	statistic	df	p.value	conf.low	conf.high
fixed	NA	(Intercept)	0.501	0.148	3.382	71	0.001	0.206	0.796
fixed	NA	Time_numeric	0.033	0.032	1.034	71	0.305	-0.031	0.098
ran_pars	Student_ID	sd__(Intercept)	0.000	NA	NA	NA	NA	NA	NA
ran_pars	Group_factor	sd__(Intercept)	0.000	NA	NA	NA	NA	NA	NA
ran_pars	Residual	sd__Observation	0.677	NA	NA	NA	NA	NA	NA

```r
# Model performance
cat("\nModel Performance:\n")
```

```
##
## Model Performance:
```

```r
performance::icc(model3)
```

```
## [1] NA
```

```r
performance::r2(model3)
```

```
## Random effect variances not available. Returned R2 does not account for random effects.
```

```
## # R2 for Mixed Models
##
##   Conditional R2: NA
##      Marginal R2: 0.015
```

Model 4: Longitudinal - Time + Co-creation

4. Does the co-creation effect vary by timepoint?

```r
# Mixed model with time and co-creation
model4 <- lmer(
  score ~ Time_numeric + Cocreation_Level + (1 | Group_factor) + (1 | Student_ID),
  data = df_long,
  REML = TRUE
)

# Summary
summary(model4)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: score ~ Time_numeric + Cocreation_Level + (1 | Group_factor) +
##     (1 | Student_ID)
##    Data: df_long
```

```
##
## REML criterion at convergence: 157.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.3858 -0.8475  0.1010  0.5617  2.3543
##
## Random effects:
##  Groups       Name        Variance Std.Dev.
##  Student_ID   (Intercept) 0.0000   0.0000
##  Group_factor (Intercept) 0.0000   0.0000
##  Residual                 0.4524   0.6726
## Number of obs: 73, groups:  Student_ID, 27; Group_factor, 9
##
## Fixed effects:
##                  Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)       0.38267    0.17040 70.00000   2.246   0.0279 *
## Time_numeric      0.03381    0.03207 70.00000   1.054   0.2953
## Cocreation_Level  0.10424    0.07579 70.00000   1.375   0.1734
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) Tm_nmr
## Time_numerc -0.735
## Cocretn_Lvl -0.504  0.010
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
# Tidy output
tidy(model4, conf.int = TRUE) %>%
  kable(digits = 3, caption = "Model 2: Time + Co-creation")
```

Table 12: Model 2: Time + Co-creation

effect	group	term	estimate	std.error	statistic	df	p.value	conf.low	conf.high
fixed	NA	(Intercept)	0.383	0.170	2.246	70	0.028	0.043	0.723
fixed	NA	Time_numeric	0.034	0.032	1.054	70	0.295	-0.030	0.098
fixed	NA	Cocreation_Level	0.104	0.076	1.375	70	0.173	-0.047	0.255
ran_pars	Student_ID	sd__(Intercept)	0.000	NA	NA	NA	NA	NA	NA
ran_pars	Group_factor	sd__(Intercept)	0.000	NA	NA	NA	NA	NA	NA
ran_pars	Residual	sd__Observation	0.673	NA	NA	NA	NA	NA	NA

```r
# Model performance
cat("\nModel Performance:\n")
```

```
##
## Model Performance:
```

```r
performance::icc(model4)
```

```
## [1] NA
```

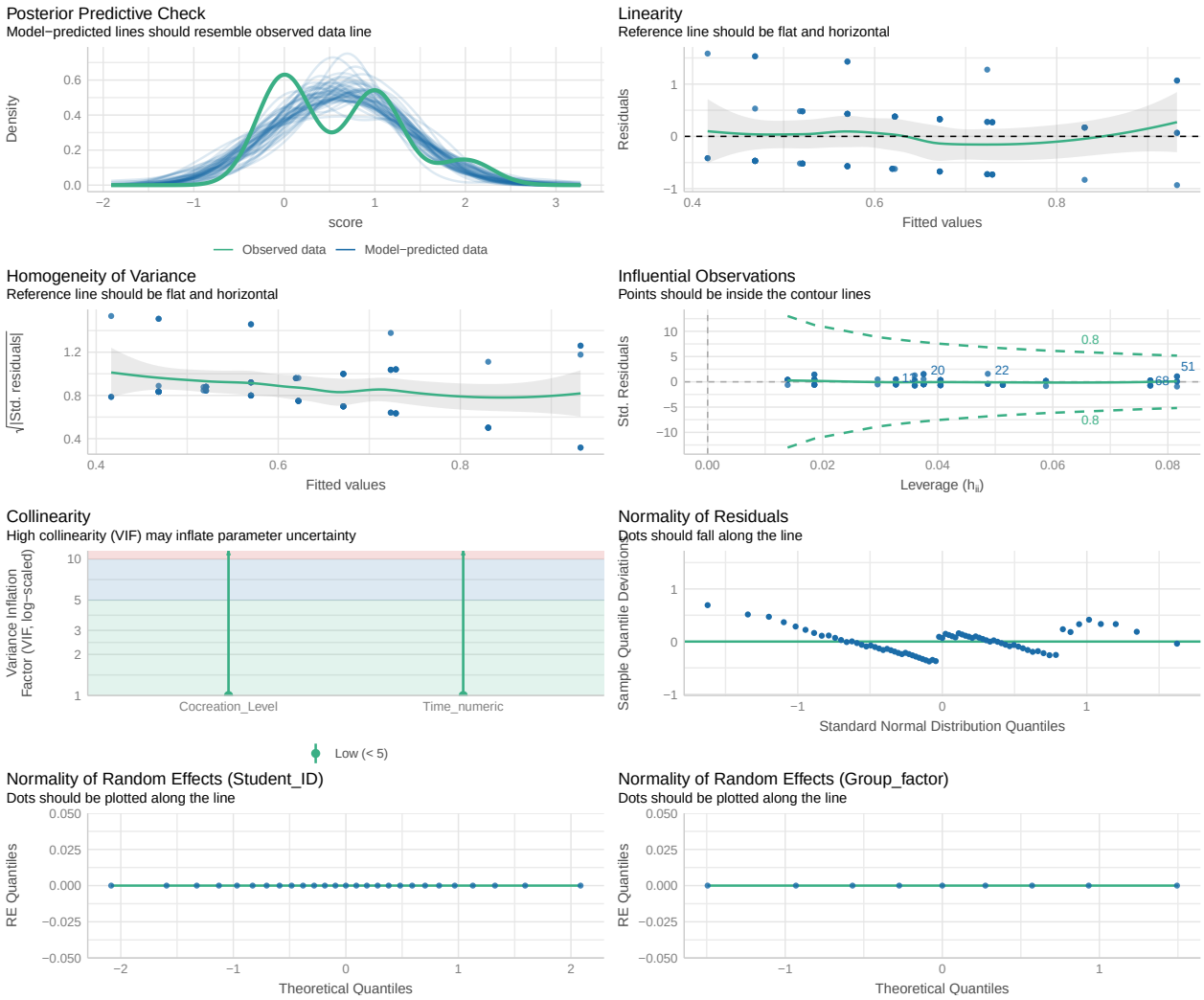
```r
performance::r2(model4)
```

```
## Random effect variances not available. Returned R2 does not account for random effects.
```

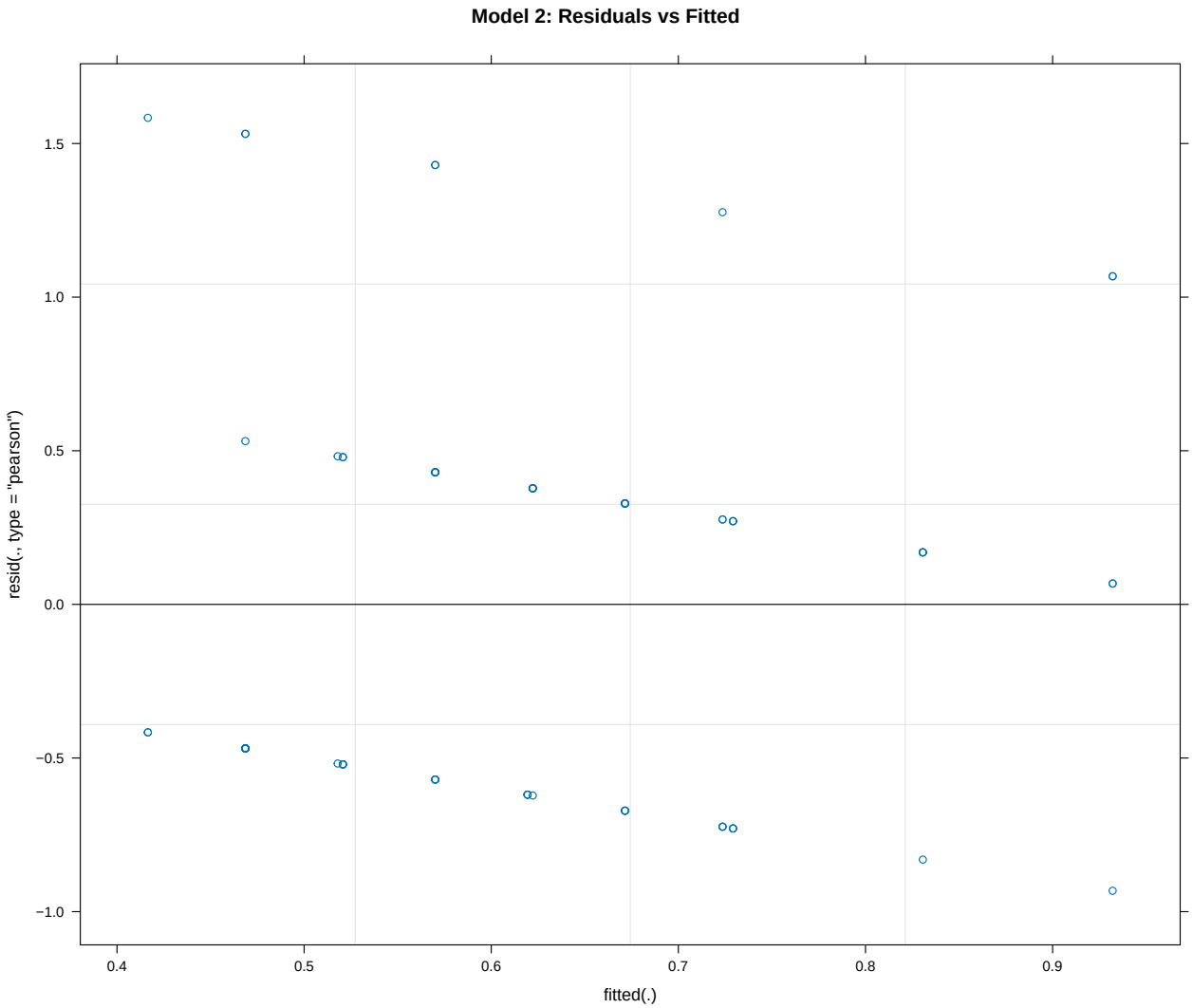
```
## # R2 for Mixed Models
##
##   Conditional R2: NA
##      Marginal R2: 0.040
```

Model 4 Diagnostics

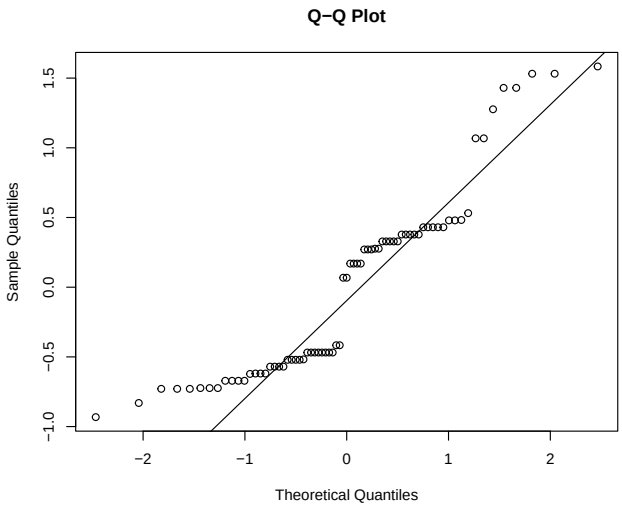
```r
check_model(model4, panel = TRUE)
```

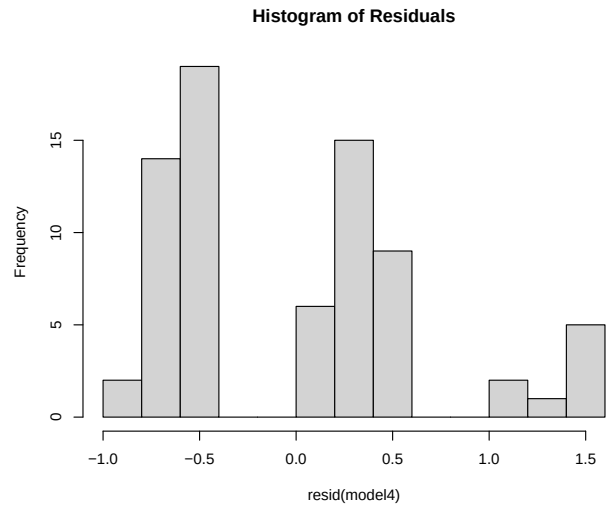


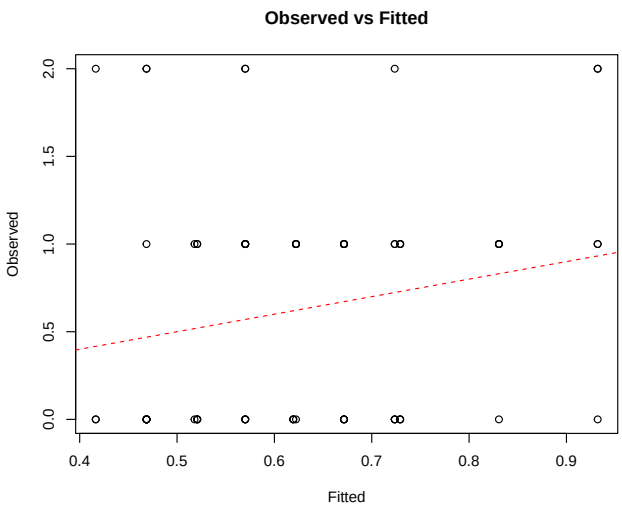

```r
# Additional diagnostics
par(mfrow = c(2, 2))
plot(model4, main = "Model 2: Residuals vs Fitted")
```




```r
qqnorm(resid(model4), main = "Q-Q Plot")
qqline(resid(model4))
hist(resid(model4), main = "Histogram of Residuals", breaks = 15)
plot(fitted(model4), df_long$score, main = "Observed vs Fitted",
     xlab = "Fitted", ylab = "Observed")
abline(0, 1, col = "red", lty = 2)
par(mfrow = c(1, 1))
```







Model 4 Interpretation

```
cat("\n=== MODEL 4 INTERPRETATION ===\n")
```

```
##
## === MODEL 4 INTERPRETATION ===
```

```
cat("Research Question: How do Time and Co-creation affect knowledge longitudinally?\n\n")
```

```
## Research Question: How do Time and Co-creation affect knowledge longitudinally?
```

```
# Extract coefficients
beta_time <- fixef(model4)["Time_numeric"]
beta_cocreation <- fixef(model4)["Cocreation_Level"]
ci_time <- confint(model4, parm = "Time_numeric", level = 0.95)
ci_cocreation <- confint(model4, parm = "Cocreation_Level", level = 0.95)
p_time <- summary(model4)$coefficients["Time_numeric", "Pr(>|t|)"]
p_cocreation <- summary(model4)$coefficients["Cocreation_Level", "Pr(>|t|)"]

cat("Fixed Effects:\n")
```

```
## Fixed Effects:
```

```r
cat(sprintf("  Time effect (beta): %.3f, 95%% CI [%.3f, %.3f], p = %.4f\n",
            beta_time, ci_time[1], ci_time[2], p_time))
```

```
##   Time effect (beta): 0.034, 95% CI [-0.029, 0.096], p = 0.2953
```

```r
cat(sprintf("  Co-creation effect (beta): %.3f, 95%% CI [%.3f, %.3f], p = %.4f\n",
            beta_cocreation, ci_cocreation[1], ci_cocreation[2], p_cocreation))
```

```
##   Co-creation effect (beta): 0.104, 95% CI [-0.043, 0.252], p = 0.1734
```

```r
cat("\nInterpretation:\n")
```

```
##
## Interpretation:
```

```r
if (p_time < 0.05) {
  if (beta_time > 0) {
    cat("  [YES] Knowledge INCREASES over time\n")
    cat(sprintf("    - %.3f points increase per timepoint unit\n", beta_time))
  } else {
    cat("  [YES] Knowledge DECREASES over time\n")
  }
} else {
  cat("  [NO] No significant time effect\n")
}
```

```
##   [NO] No significant time effect
```

```r
if (p_cocreation < 0.05) {
  cat("  [YES] Co-creation predicts knowledge across all timepoints\n")
  cat(sprintf("    - %.3f points per co-creation level\n", beta_cocreation))
} else {
  cat("  [NO] No significant co-creation effect when accounting for time\n")
}
```

```
##   [NO] No significant co-creation effect when accounting for time
```

```r
# Random effects
var_group <- as.data.frame(VarCorr(model4))$vcov[1]
var_student <- as.data.frame(VarCorr(model4))$vcov[2]
var_resid <- attr(VarCorr(model4), "sc")^2

cat("\nRandom Effects (Variance Components):\n")
```

```
##
## Random Effects (Variance Components):
```

```r
cat(sprintf("  Group level: %.3f\n", var_group))
```

```
##   Group level: 0.000
```

```r
cat(sprintf("  Student level: %.3f\n", var_student))
```

```
##   Student level: 0.000
```

```r
cat(sprintf("  Residual: %.3f\n", var_resid))
```

```
##   Residual: 0.452
```

```r
total_var <- var_group + var_student + var_resid
cat(sprintf("\nVariance Partitioning:\n"))
```

```
##
## Variance Partitioning:
```

```r
cat(sprintf("  Between groups: %.1f%%\n", var_group/total_var*100))
```

```
##   Between groups: 0.0%
```

```r
cat(sprintf("  Between students (within groups): %.1f%%\n", var_student/total_var*100))
```

```
##   Between students (within groups): 0.0%
```

```r
cat(sprintf("  Residual (within students): %.1f%%\n", var_resid/total_var*100))
```

```
##   Residual (within students): 100.0%
```

```r
# Model fit
r2_vals <- r2(model4)
```

```
## Random effect variances not available. Returned R2 does not account for random effects.
```

```r
cat(sprintf("\nModel Fit:\n"))
```

```
##
## Model Fit:
```

```r
cat(sprintf("  Marginal R²: %.3f (variance explained by fixed effects)\n", r2_vals$R2_marginal))
```

```
##   Marginal R²: 0.040 (variance explained by fixed effects)
```

```r
cat(sprintf("  Conditional R²: %.3f (variance explained by full model)\n", r2_vals$R2_conditional))
```

```
##   Conditional R²: NA (variance explained by full model)
```

```r
cat(sprintf("  AIC: %.1f, BIC: %.1f\n", AIC(model4), BIC(model4)))
```

```
##   AIC: 169.9, BIC: 183.6
```

```r
cat("\n================================\n")
```

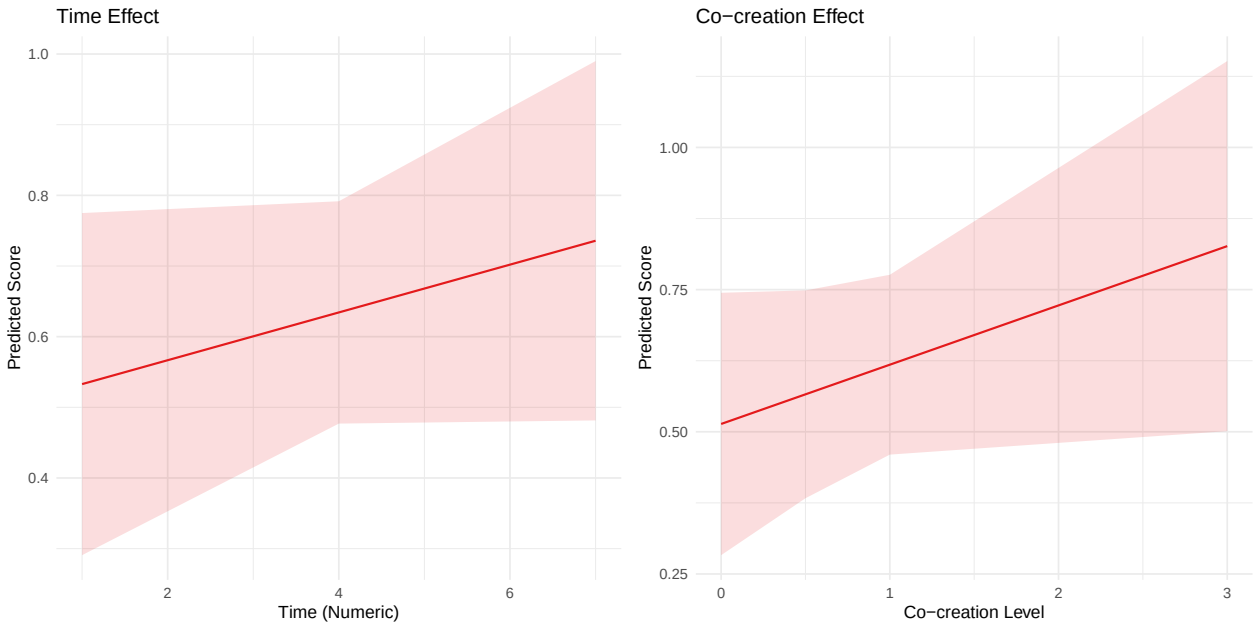
```
##
## ================================
```

Model 4 Visualization

```r
# Plot both effects
p1 <- plot_model(model4, type = "pred", terms = "Time_numeric") +
  labs(title = "Time Effect", x = "Time (Numeric)", y = "Predicted Score") +
  theme_minimal(base_size = 12)

p2 <- plot_model(model4, type = "pred", terms = "Cocreation_Level") +
  labs(title = "Co-creation Effect", x = "Co-creation Level", y = "Predicted Score") +
  theme_minimal(base_size = 12)

grid.arrange(p1, p2, ncol = 2)
```




```r
# ============================================
# MODEL 4: LONGITUDINAL ANALYSIS (SIMPLER APPROACH)
# ============================================

library(sandwich)
library(lmtest)
library(broom)
library(knitr)

# Fit simple linear regression with time and co-creation
model4_simple <- lm(score ~ Time_numeric + Cocreation_Level, data = df_long)

# Get cluster-robust standard errors (clustered by Student_ID for repeated measures)
model4_robust <- coeftest(model4_simple,
                          vcov = vcovCL,
                          cluster = ~Student_ID)

# Display results
cat("MODEL 4: Longitudinal Analysis - Time + Co-creation Effects\n")
```

```
## MODEL 4: Longitudinal Analysis - Time + Co-creation Effects
```

```r
cat("============================================================\n")
```

```
## ============================================================
```

```r
cat("Cluster-robust SEs accounting for repeated measures\n\n")
```

```
## Cluster-robust SEs accounting for repeated measures
```

```r
print(model4_robust)
```

```
##
## t test of coefficients:
##
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.382672   0.187526  2.0406  0.04506 *
```

```
## Time_numeric     0.033815   0.038030  0.8891  0.37697
## Cocreation_Level 0.104238   0.044788  2.3274  0.02284 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Get confidence intervals with robust SEs
robust_ci_model4 <- coefci(model4_simple,
                           vcov = vcovCL,
                           cluster = ~Student_ID)
cat("\nConfidence Intervals:\n")
```

```
##
## Confidence Intervals:
```

```r
print(robust_ci_model4)
```

```
##                         2.5 %    97.5 %
## (Intercept)       0.008662156 0.7566815
## Time_numeric     -0.042034343 0.1096635
## Cocreation_Level  0.014912074 0.1935647
```

```r
# Create a tidy table
robust_results_model4 <- tidy(model4_robust, conf.int = TRUE) %>%
  kable(digits = 3, caption = "Model 4: Longitudinal Analysis with Cluster-Robust SE")

print(robust_results_model4)
```

```
##
##
## Table: Model 4: Longitudinal Analysis with Cluster-Robust SE
##
## |term             | estimate| std.error| statistic| p.value| conf.low| conf.high|
## |:----------------|--------:|---------:|---------:|-------:|--------:|---------:|
## |(Intercept)      |    0.383|     0.188|     2.041|   0.045|    0.009|     0.757|
## |Time_numeric     |    0.034|     0.038|     0.889|   0.377|   -0.042|     0.110|
## |Cocreation_Level |    0.104|     0.045|     2.327|   0.023|    0.015|     0.194|
```

```r
# Model fit statistics
cat("\n=== Model 4 Fit Statistics ===\n")
```

```
##
## === Model 4 Fit Statistics ===
```

```r
cat("R-squared:", summary(model4_simple)$r.squared, "\n")
```

```
## R-squared: 0.04074072
```

```r
cat("Adjusted R-squared:", summary(model4_simple)$adj.r.squared, "\n")
```

```
## Adjusted R-squared: 0.01333331
```

```r
cat("\nNumber of observations:", nobs(model4_simple), "\n")
```

```
##
## Number of observations: 73
```

```r
cat("Number of students (clusters):", length(unique(df_long$Student_ID)), "\n")
```

```
## Number of students (clusters): 27
```

```r
cat("Number of groups:", length(unique(df_long$Group_factor)), "\n")
```

```
## Number of groups: 9
```

```r
# Interpretation guide
cat("\n=== Model 4 Interpretation ===\n")
```

```
##
## === Model 4 Interpretation ===
```

```r
cat("Time_numeric: Change in score per timepoint unit (learning trajectory)\n")
```

```
## Time_numeric: Change in score per timepoint unit (learning trajectory)
```

```r
cat("Cocreation_Level: Effect of co-creation level across all timepoints\n")
```

```
## Cocreation_Level: Effect of co-creation level across all timepoints
```

Model 5: Time × Co-creation Interaction

Does the co-creation effect differ by timepoint?

```r
# Mixed model with interaction
model5 <- lmer(
  score ~ Time_numeric * Cocreation_Level + (1 | Group_factor) + (1 | Student_ID),
  data = df_long,
  REML = TRUE
)

# Summary
summary(model5)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: score ~ Time_numeric * Cocreation_Level + (1 | Group_factor) +
##     (1 | Student_ID)
##    Data: df_long
##
## REML criterion at convergence: 160
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.8771 -0.8479 -0.3689  0.6543  2.1684
##
## Random effects:
##  Groups       Name        Variance Std.Dev.
##  Student_ID   (Intercept) 0.0000   0.0000
##  Group_factor (Intercept) 0.0000   0.0000
##  Residual                 0.4397   0.6631
## Number of obs: 73, groups:  Student_ID, 27; Group_factor, 9
##
## Fixed effects:
##                              Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                   0.60918    0.21250 69.00000   2.867  0.00549 **
## Time_numeric                 -0.02487    0.04622 69.00000  -0.538  0.59231
## Cocreation_Level             -0.09653    0.13744 69.00000  -0.702  0.48481
## Time_numeric:Cocreation_Level  0.05234    0.03007 69.00000   1.740  0.08623 .
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) Tm_nmr Ccrt_L
## Time_numerc -0.844
## Cocretn_Lvl -0.731  0.616
## Tm_nmrc:C_L  0.612 -0.729 -0.839
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# Tidy output
tidy(model5, conf.int = TRUE) %>%
  kable(digits = 3, caption = "Model 5: Time × Co-creation Interaction")
```

Table 13: Model 5: Time × Co-creation Interaction

effect	group	term	estimate	std.error	statistic	df	p.value	conf.low	conf.high
fixed	NA	(Intercept)	0.609	0.212	2.867	69	0.005	0.185	1.033
fixed	NA	Time_numeric	-0.025	0.046	-0.538	69	0.592	-0.117	0.067
fixed	NA	Cocreation_Level	-0.097	0.137	-0.702	69	0.485	-0.371	0.178
fixed	NA	Time_numeric:Cocreation_Level	0.052	0.030	1.740	69	0.086	-0.008	0.112
ran_pars	Student_ID	sd__(Intercept)	0.000	NA	NA	NA	NA	NA	NA
ran_pars	Group_factor	sd__(Intercept)	0.000	NA	NA	NA	NA	NA	NA
ran_pars	Residual	sd__Observation	0.663	NA	NA	NA	NA	NA	NA

```
# Model performance
cat("\nModel Performance:\n")
```

```
##
## Model Performance:
```

```
performance::icc(model5)
```

```
## [1] NA
```

```
performance::r2(model5)
```

```
## Random effect variances not available. Returned R2 does not account for random effects.
```

```
## # R2 for Mixed Models
##
##   Conditional R2: NA
##      Marginal R2: 0.078
```

```
# Test significance of interaction
cat("\nInteraction test:\n")
```

```
##
## Interaction test:
```

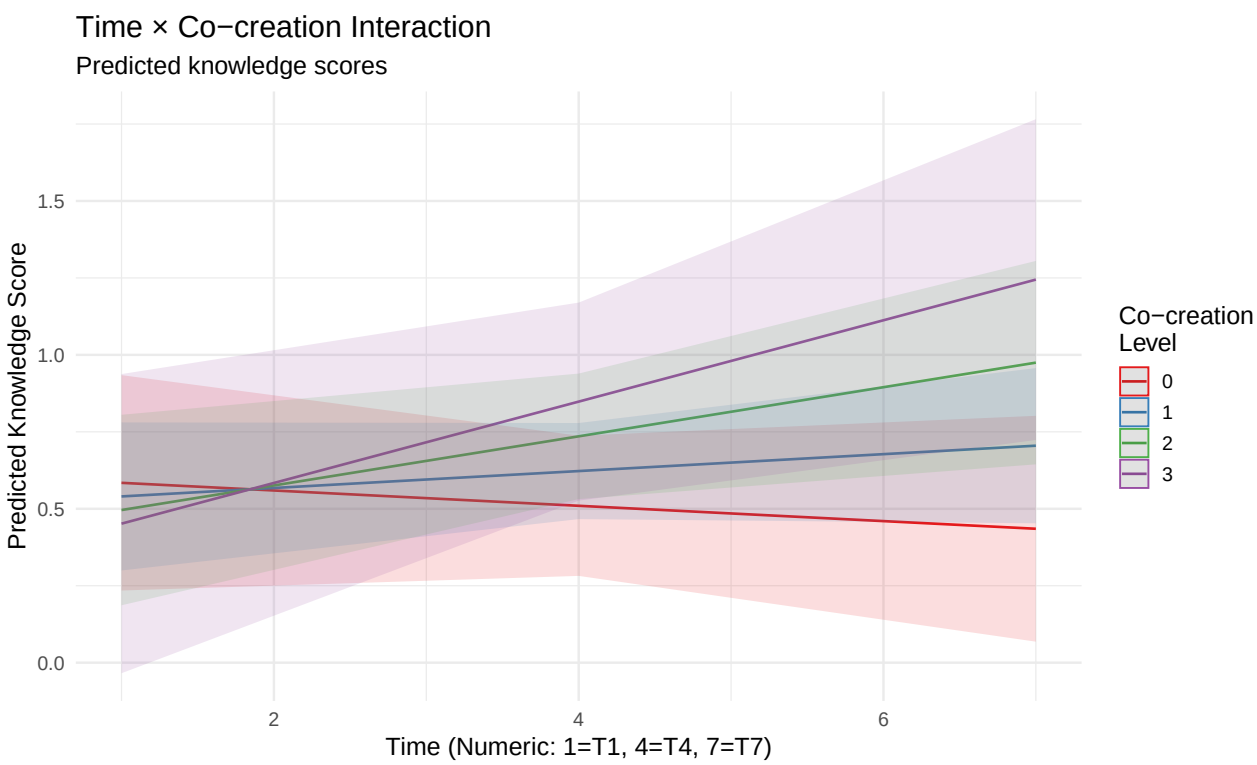
```
anova(model4, model5)
```

```
## Data: df_long
## Models:
## model4: score ~ Time_numeric + Cocreation_Level + (1 | Group_factor) + (1 | Student_ID)
## model5: score ~ Time_numeric * Cocreation_Level + (1 | Group_factor) + (1 | Student_ID)
```

```
##        npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
## model4    6 158.20 171.94 -73.100   146.20
## model5    7 157.06 173.10 -71.532   143.06 3.1366  1    0.07656 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Model 5 Visualization

```r
# Plot interaction
plot_model(model5, type = "pred", terms = c("Time_numeric", "Cocreation_Level [0, 1, 2, 3]")) +
  labs(
    title = "Time × Co-creation Interaction",
    subtitle = "Predicted knowledge scores",
    x = "Time (Numeric: 1=T1, 4=T4, 7=T7)",
    y = "Predicted Knowledge Score",
    color = "Co-creation\nLevel"
  ) +
  theme_minimal(base_size = 14)
```



Model 5 Interpretation

```r
cat("\n=== MODEL 5 INTERPRETATION ===\n")
```

```
##
## === MODEL 5 INTERPRETATION ===
```

```r
cat("Research Question: Does the co-creation effect change over time?\n\n")
```

```
## Research Question: Does the co-creation effect change over time?
```

```r
# Extract coefficients
beta_interaction <- fixef(model5)["Time_numeric:Cocreation_Level"]
ci_interaction <- confint(model5, parm = "Time_numeric:Cocreation_Level", level = 0.95)
p_interaction <- summary(model5)$coefficients["Time_numeric:Cocreation_Level", "Pr(>|t|)"]

cat("Interaction Effect:\n")
```

```
## Interaction Effect:
```

```r
cat(sprintf("  Time x Co-creation (beta): %.3f\n", beta_interaction))
```

```
##   Time x Co-creation (beta): 0.052
```

```r
cat(sprintf("  95%% CI: [%.3f, %.3f]\n", ci_interaction[1], ci_interaction[2]))
```

```
##   95% CI: [-0.006, 0.110]
```

```r
cat(sprintf("  p-value: %.4f\n", p_interaction))
```

```
##   p-value: 0.0862
```

```r
cat("\nInterpretation:\n")
```

```
##
## Interpretation:
```

```r
if (p_interaction < 0.05) {
  cat("  [YES] SIGNIFICANT INTERACTION\n")
  if (beta_interaction > 0) {
    cat("  -> The co-creation effect INCREASES over time\n")
    cat("  -> Higher co-creation levels show greater learning gains from T1 to T7\n")
    cat("  -> This suggests co-creation has a cumulative benefit\n")
  } else {
    cat("  -> The co-creation effect DECREASES over time\n")
    cat("  -> Benefits of co-creation diminish at later timepoints\n")
  }
} else {
  cat("  [NO] NO SIGNIFICANT INTERACTION\n")
  cat("  -> The co-creation effect is CONSISTENT across timepoints\n")
  cat("  -> Co-creation has the same impact at T1, T4, and T7\n")
}
```

```
##   [NO] NO SIGNIFICANT INTERACTION
##   -> The co-creation effect is CONSISTENT across timepoints
##   -> Co-creation has the same impact at T1, T4, and T7
```

```r
# Model comparison
cat("\n Model Comparison:\n")
```

```
##
##  Model Comparison:
```

```r
anova_result <- anova(model4, model5)
p_lr_test <- anova_result$`Pr(>Chisq)`[2]
cat(sprintf("  Likelihood ratio test: chi-squared(1) = %.2f, p = %.4f\n",
            anova_result$Chisq[2], p_lr_test))
```

```
##   Likelihood ratio test: chi-squared(1) = 3.14, p = 0.0766
```

```r
if (p_lr_test < 0.05) {
  cat("  -> Model with interaction fits significantly better\n")
} else {
  cat("  -> Interaction does not significantly improve model fit\n")
  cat("  -> Simpler model (Model 4) is preferred\n")
}
```

```
##   -> Interaction does not significantly improve model fit
##   -> Simpler model (Model 4) is preferred
```

```r
# Effect sizes
r2_vals <- r2(model5)
```

```
## Random effect variances not available. Returned R2 does not account for random effects.
```

```r
cat(sprintf("\nModel Fit:\n"))
```

```
##
## Model Fit:
```

```r
cat(sprintf("  Marginal R²: %.3f\n", r2_vals$R2_marginal))
```

```
##   Marginal R²: 0.078
```

```r
cat(sprintf("  Conditional R²: %.3f\n", r2_vals$R2_conditional))
```

```
##   Conditional R²: NA
```

```r
cat("\n================================\n")
```

```
##
## ================================
```

Model 6: T4 Predicts Co-creation (Selection Effect)?

Additional question:Does previous knowledge (T4) predict co-creation level (selection effect)?

```r
# Filter T1 data
df_t4 <- df_long %>% filter(Timepoint == "T4")

# Since co-creation is at group level, we need group-level analysis
df_t4_group <- df_t4 %>%
  group_by(Group_factor) %>%
  summarise(
    mean_T4_score = mean(score, na.rm = TRUE),
    Cocreation_Level = first(Cocreation_Level)
  )

# Simple linear model (no random effects needed - one obs per group)
model6 <- lm(Cocreation_Level ~ mean_T4_score, data = df_t4_group)

# Summary
summary(model6)
```

```
##
## Call:
## lm(formula = Cocreation_Level ~ mean_T4_score, data = df_t4_group)
##
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
## -0.98421 -0.72895 -0.23947 -0.06579  1.93421
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)     0.7395     0.9095   0.813    0.443
## mean_T4_score   0.4895     1.0818   0.452    0.665
##
## Residual standard error: 1.172 on 7 degrees of freedom
## Multiple R-squared:  0.02842,    Adjusted R-squared:  -0.1104
## F-statistic: 0.2047 on 1 and 7 DF,  p-value: 0.6646
```

```r
# Tidy output
tidy(model6, conf.int = TRUE) %>%
  kable(digits = 3, caption = "Model 2: T1 Knowledge → Co-creation Level")
```

Table 14: Model 2: T1 Knowledge → Co-creation Level

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.739	0.909	0.813	0.443	-1.411	2.890
mean_T4_score	0.489	1.082	0.452	0.665	-2.069	3.048

Model Comparison

```r
# Compare models
comparison <- compare_performance(model1, model2, model3, model4, model5, verbose = FALSE)
```

```
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
```

```r
comparison %>%
  kable(digits = 3, caption = "Model Comparison")
```

Table 15: Model Comparison

Name	Model	AIC	AIC_wt	AICc	AICc_wt	BIC	BIC_wt	RMSE	Sigma	R2_conditional	R2_marginal	R2	R2_adjusted
model1	lmerModLmerTest	51.012	0	53.234	0	55.554	0	0.616	0.645	NA	0.213	NA	NA
model2	lm	32.387	1	37.187	1	32.979	1	1.048	1.188	NA	NA	0	-0.143
model3	lmerModLmerTest	157.147	0	159.042	0	169.599	0	0.667	0.677	NA	0.015	NA	NA
model4	lmerModLmerTest	157.200	0	159.473	0	171.943	0	0.659	0.673	NA	0.040	NA	NA
model5	lmerModLmerTest	157.064	0	158.787	0	173.097	0	0.645	0.663	NA	0.078	NA	NA

```r
# AIC/BIC comparison
cat("\nLower AIC/BIC indicates better model fit\n")
```

```
##
## Lower AIC/BIC indicates better model fit
```

Summary of Results

```r
# Create summary table
results_summary <- tibble(
  Model = c(
    "Model 1: Co-creation predicting T7",
    "Model 2: T1 predicting Co-creation",
    "Model 3: Longitudinal",
    "Model 4: Longitudinal with co-creation",
    "Model 5: Interaction"
  ),
  Formula = c(
    "Score ~ Co-creation + (1|Group)",
    "Co-creation ~ T1_Score",
    "Score ~ Time + (1|Group) + (1|Student)",
    "Score ~ Time + Co-creation + (1|Group) + (1|Student)",
    "Score ~ Time × Co-creation + (1|Group) + (1|Student)"
  ),

  Key_Finding = c(
  # Model 1: mixed model (T7 only)
  paste0("Co-creation beta = ", round(fixef(model1)[2], 3),
         ", p = ", format.pval(coef(summary(model1))[2, 5], digits = 3)),
  # Model 2: simple lm (T1 -> Co-creation)
  paste0("T1 beta = ", round(coef(summary(model2))[2, 1], 3),
         ", p = ", format.pval(coef(summary(model2))[2, 4], digits = 3)),
  # Model 3: mixed model (time only)
  paste0("Time beta = ", round(fixef(model3)[2], 3),
         ", p = ", format.pval(coef(summary(model3))[2, 5], digits = 3)),
  # Model 4: mixed model (time + co-creation)
  paste0("Co-creation beta = ", round(fixef(model4)[3], 3),
         ", p = ", format.pval(coef(summary(model4))[3, 5], digits = 3)),
  # Model 5: mixed model (interaction)
  paste0("Interaction beta = ", round(fixef(model5)[4], 3),
         ", p = ", format.pval(coef(summary(model5))[4, 5], digits = 3))
)
)

results_summary %>%
  kable(caption = "Summary of All Models")
```

Table 16: Summary of All Models

Model	Formula	Key_Finding
Model 1: Co-creation predicting T7	Score ~ Co-creation + (1|Group)	Co-creation beta = 0.315, p = 0.0236
Model 2: T1 predicting Co-creation	Co-creation ~ T1_Score	T1 beta = 0.025, p = 0.976
Model 3: Longitudinal	Score ~ Time + (1|Group) + (1|Student)	Time beta = 0.033, p = 0.305
Model 4: Longitudinal with co-creation	Score ~ Time + Co-creation + (1|Group) + (1|Student)	Co-creation beta = 0.104, p = 0.173

Model	Formula	Key_Finding
Model 5: Interaction	Score ~ Time × Co-creation + (1|Group) + (1|Student)	Interaction beta = 0.052, p = 0.0862

Interpretation

Key Findings

```
# Extract key statistics
t7_effect <- fixef(model1)["Cocreation_Level"]
t7_p <- summary(model1)$coefficients["Cocreation_Level", "Pr(>|t|)"]

long_effect <- fixef(model4)["Cocreation_Level"]
long_p <- summary(model4)$coefficients["Cocreation_Level", "Pr(>|t|)"]

t1_effect <- coef(model2)["mean_T1_score"]
t1_p <- summary(model2)$coefficients["mean_T1_score", "Pr(>|t|)"]

cat("=== INTERPRETATION ===\n\n")
```

```
## === INTERPRETATION ===
```

```
cat("1. CO-CREATION EFFECT ON LEARNING (T7):\n")
```

```
## 1. CO-CREATION EFFECT ON LEARNING (T7):
```

```
cat(sprintf("   beta = %.3f, p = %.4f\n", t7_effect, t7_p))
```

```
##    beta = 0.315, p = 0.0236
```

```
if (t7_p < 0.05) {
  cat("   [YES] SIGNIFICANT: Higher co-creation predicts better post-test knowledge\n")
} else {
  cat("   [NO] NOT SIGNIFICANT\n")
}
```

```
##    [YES] SIGNIFICANT: Higher co-creation predicts better post-test knowledge
```

```
cat("\n2. LONGITUDINAL CO-CREATION EFFECT:\n")
```

```
##
## 2. LONGITUDINAL CO-CREATION EFFECT:
```

```
cat(sprintf("   beta = %.3f, p = %.4f\n", long_effect, long_p))
```

```
##    beta = 0.104, p = 0.1734
```

```
if (long_p < 0.05) {
  cat("   [YES] SIGNIFICANT: Co-creation effect holds across all timepoints\n")
} else {
  cat("   [NO] NOT SIGNIFICANT: Effect may be specific to T7\n")
}
```

```
##    [NO] NOT SIGNIFICANT: Effect may be specific to T7
```

```
cat("\n3. SELECTION EFFECT (T1 -> Co-creation):\n")
```

```
##
```

```
## 3. SELECTION EFFECT (T1 -> Co-creation):
```

```r
cat(sprintf("   beta = %.3f, p = %.4f\n", t1_effect, t1_p))
```

```
##    beta = 0.025, p = 0.9759
```

```r
if (t1_p < 0.05) {
  cat("   [WARNING] SIGNIFICANT: Baseline knowledge predicts co-creation level\n")
  cat("   This suggests potential selection effects\n")
} else {
  cat("   [OK] NOT SIGNIFICANT: No evidence of selection bias\n")
  cat("   Co-creation appears independent of baseline knowledge\n")
}
```

```
##    [OK] NOT SIGNIFICANT: No evidence of selection bias
##    Co-creation appears independent of baseline knowledge
```

Recommended Reporting

```r
cat("\n=== SUGGESTED TEXT FOR PAPER ===\n\n")
```

```
##
## === SUGGESTED TEXT FOR PAPER ===
```

```r
cat("We examined the relationship between co-creation level during LLM interaction\n")
```

```
## We examined the relationship between co-creation level during LLM interaction
```

```r
cat("and learning outcomes using mixed-effects models to account for students nested\n")
```

```
## and learning outcomes using mixed-effects models to account for students nested
```

```r
cat("within groups.\n\n")
```

```
## within groups.
```

```r
if (t7_p < 0.05) {
  cat(sprintf("Co-creation level significantly predicted post-test knowledge (T7)\n"))
  cat(sprintf("(beta = %.2f, 95%% CI [%.2f, %.2f], p = %.3f).\n",
              t7_effect,
              confint(model1)["Cocreation_Level", 1],
              confint(model1)["Cocreation_Level", 2],
              t7_p))
}
```

```
## Co-creation level significantly predicted post-test knowledge (T7)
## (beta = 0.32, 95% CI [0.06, 0.57], p = 0.024).
```

```r
cat("\nThe mixed model showed a singular fit with zero group-level variance,\n")
```

```
##
## The mixed model showed a singular fit with zero group-level variance,
```

```r
cat("indicating that variability in knowledge scores is primarily within-group\n")
```

```
## indicating that variability in knowledge scores is primarily within-group
```

```r
cat("rather than between-group differences.\n\n")
```

```
## rather than between-group differences.
```

```r
if (t1_p >= 0.05) {
  cat("\nImportantly, baseline knowledge (T1) did not predict co-creation level\n")
  cat(sprintf("(beta = %.2f, p = %.3f), suggesting that the observed co-creation effect\n",
              t1_effect, t1_p))
  cat("represents a genuine learning effect rather than a selection artifact.\n")
}
```

```
##
## Importantly, baseline knowledge (T1) did not predict co-creation level
## (beta = 0.02, p = 0.976), suggesting that the observed co-creation effect
## represents a genuine learning effect rather than a selection artifact.
```